MOBILE LOGISTICS ROBOT

PROJECT REPORT

Submitted by

KIRUBAKARAN P	**19DM23**
MOHAMED TARIQ M	**19DM25**
MULLAINATHAN V H	**19DM27**
UDHAYA KUMAR T	**19DM53**

Under the guidance of

Mr. R. ANGURAJ

In partial fulfillment of the requirement for the award of

DIPLOMA IN MECHATRONICS

STATE BOARD OF TECHNICAL EDUCATION

GOVERNMENT OF TAMILNADU

JUNE 2022

DEPARTMENT OF MECHATRONICS

PSG POLYTECHNIC COLLEGE

(Autonomous and an ISO 9001:2015 certified Institution)

COIMBATORE – 641 004

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CERTIFICATE

This is to certify that the Project report entitled

MOBILE LOGISTICS ROBOT

has been submitted by

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MULLAINATHAN V H	**19DM27**
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In partial fulfillment for the award of

DIPLOMA IN MECHATRONICS

of the State Board of Technical Education,

Government of Tamil Nadu.

during the academic year **2021-2022**

……..……	**……..**
Faculty guide	Head of the Department

Certified that the candidate was examined by us in the Project work viva-voce examination held on **……**

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To Whom It May Concern,

This is to certify that the following students from PSG Polytechnic College have done an internship in our organization from October 2021 to May 2022.

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His performance has been excellent throughout the internship and his conduct was also very good. We wish him good luck in his future endeavors.

Best Regards,

For THEDUSH ROBOTICS PRIVATE LIMITED
DIRECTOR

Thedush Robotics Private Limited.

ACKNOWLEDGEMENT

First of all we extend our sincere thanks to **Dr.B.Giriraj**, Principal, PSG Polytechnic College for having provided mini project work and viva voce as a part of our curriculum and given us this opportunity to undergo the same.

We extend Our sincere thanks to **Mrs.D.Sujatha**, Head of the Department, Department of Mechatronics who granted the permission to proceed with the mini project work.

We would like to extend our profound thanks and regards to **Mr. Prajeeth Kumar**, Founder and his friend **Mr. Navaneeth**, **M/s Thedush Robotics** for providing the Resources to work in their Project.

We take this opportunity to express my profound gratitude to our Project guide **Mr.R.Anguraj**, Faculty guide, for his guidance, monitoring and constant encouragement throughout the course of work and for his support to complete the report.

We are obliged to the staff members of Department of Mechatronics, for the valuable support provided by them in their respective fields. We are grateful for their co-operation during the period of the Project.

Any omission in this brief acknowledgement does not mean lack of gratitude.

ABSTRACT

Introducing worker robots into warehouses has the potential to increase the standards currently held by the industry. The autonomous machines have the capability of dynamically storing pallets, lowering labor and insurance rates, and reducing the risk of personal injury. To broach these goals, the Automated Guided Vehicle, AGV, was designed to collaborate with another autonomous vehicle, the Automated Storage and Retrieval System, ASRS.

Although they appear similar, the AGV and ASRS serve two separate purposes. Catered to industries whose products can expire, the ASRS keeps track of when each pallet enters the warehouse and its current location. This ensures that the oldest product gets shipped out first. The ASRS is a tall vehicle designed to navigate in the tight aisles of a warehouse floor. Although its tall mast gives it the capability of lifting pallets onto a third tier shelf, it restricts the vehicle travels to slow speeds to maintain a safe environment. With all considerations in mind, the ASRS cannot efficiently traverse long distances in a warehouse. The problem is easily solved with the introduction of a small, light vehicle. The AGV can quickly travel across a warehouse while carrying a pallet. Its simple pallet jack can lift up and set down pallets at one height. Because of its small body and simple forklift mechanism, the AGV is much less expensive than the ASRS.

The AGV receives tasks from the ASRS, which can vary from picking up a pallet at the incoming dock to delivering one a pallet to the outgoing dock. Accurate navigation is obtained by mapping high contrast lines on the warehouse floor in software. While navigating and backing up, the vehicle senses its surrounding to predict an imminent collision.

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CHAPTER 1
INTRODUCTION

CHAPTER 1

INTRODUCTION

1.1 INTRODUCTION TO AGV:

Advancements in manufacturing technology allow companies to rapidly produce products. This has provoked a trend to reduce bulk inventory in favour of short-term supplies. Although this allows corporations more financial freedom, it requires warehouses to accommodate temporary, selective storage. Improved product handling and speed can be achieved with the implementation of an Automated Guided Vehicle, AGV.

In a traditional warehouse, human safety governs the productivity. With the help of intelligent computers, the AGV can safely achieve higher speeds. Precision turning allows it to accurately navigate in tight spaces. The AGV is highly flexible as a result of remote communication. Its ability to communicate with other autonomous vehicles provides a seamless operation. Continuous coordination between vehicles delivers money saving efficiency.



Fig 1.1 AGV in Shop Floor with Robotic Arm attachment

The introduction of unmanned vehicles onto a warehouse floor has favorable effects on safety. With the aid of environmental sensors, the AGV can detect objects in its collision path. Automation eliminates vehicle traffic jams and their potential for accidents.

For companies building new warehouses, there are many monetary benefits to investing in intelligent machinery. The workforce required to run the warehouse and the additional overhead (e.g., insurance) required to support that overhead will be drastically reduced. Increased product turn-around and faster shipping will result in more satisfied customers. Also, automation reduces the risks of personal injury.

1.2 NAVIGATION:

Automated Guided Vehicles (AGVs) are also known by other names such as LGV (Laser-Guided Vehicle), Mobile Robots, SGV (Self-Guided Vehicle), Guided Carts, Autonomous Vehicles, and Driverless Vehicles.

When developing or upgrading an automated guided vehicle (AGV) or mobile robot, investing in the right autonomous navigation technology is key. Choosing the wrong AGV navigation method can have a dramatic effect on the cost, efficiency and therefore success of the automated solution you produce.

- ✓ Navigation technologies that guide vehicles along physical lines (a.k.a. line following). These lines could take the form of magnetic tape, painted lines, inductive wire or (less often these days) rails. We can also include tags in this category for simplicity.
- ✓ Navigation technologies that guide vehicles along virtual (i.e. digital) paths, which exist only in the vehicle's software.

1.2.1 LINE FOLLOWING NAVIGATION:

With line following navigation technologies, AGVs are literally guided through a facility by a physical line, such as magnetic tape, inductive wire (installed in/under the floor) or painted lines.

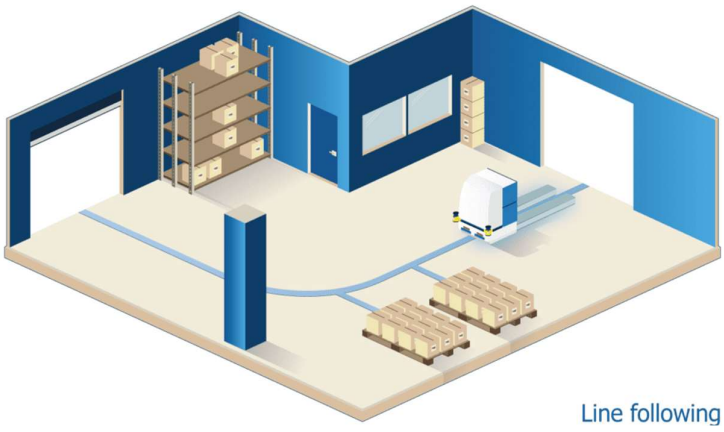


Fig 1.2 Line Following Navigation

This approach works in the following way: the vehicle features a sensor that 'looks' down at the line on the floor. Then, it measures the left and right error, and uses this information to correct the vehicle's trajectory.

This simple AGV navigation method is ideal for applications in which a vehicle moves continually around a loop. A common example is automotive manufacturing, where mobile robots move parts from station to station.

On the plus side then, line following is simple, exceptionally reliable, accurate and stable. However, it does have its drawbacks:

- ✓ Firstly, it takes time to layout and, in future, modify routes. This is especially true in the case of inductive wire, which is literally installed in the floor itself.
- ✓ Managing a fleet of line-following vehicles can be difficult if not impossible, especially when several lines need to cross each other. Traffic management can also be extremely difficult for complex installations.
- ✓ The lines themselves can also require maintenance. Magnetic tape, for example, can get ripped up over time.

1.2.2 TAG FOLLOWING NAVIGATION:

Tag following works in much the same way as the following of physical lines. However, in this case AGVs are guided through a facility by tags such as QR codes, RFID tags or magnetic points embedded in the floor.



Fig 1.3 Tag Following Navigation

Whatever the type of tag, the vehicle is equipped with a specific sensor to follow the route defined by these tags – in that sense this approach is just like line following. Tag following is the reference in e-commerce, where simple path layouts are the norm.

The pros and cons of tag navigation are therefore very similar to those of line following technologies: tags take time to physically install (and modify in future), and maintenance will be required over time, however, tags are exceptionally reliable and robust in operation.

1.2.3 LASER TRIANGULATION:

The working principle of laser triangulation (sometimes called laser target navigation or simply laser navigation) is similar to GPS navigation.

Laser triangulation uses a minimum of three references to triangulate the position of the vehicle. Unlike GPS however, which uses satellites, this methodology utilizes a laser scanner that sits on top of the vehicle. Its beams are reflected off reflective targets that are professionally installed in the environment.



Fig 1.4 Laser Triangulation Navigation

Unlike line and tag following AGVs, where the logic of the mission is described by what is on the floor, with triangulation the logic is instead programmed using software.

As with line-following, laser triangulation technology is exceptionally reliable. It is capable of high positioning accuracy (required for efficient picking, dropping, and vehicle charging), and relatively fast vehicle speeds are possible.

What's more, since this technology uses virtual paths instead of physical ones, maintenance costs are lower and effective fleet and traffic management are much easier than with line following and tags. Route modifications are also quick and simple – unless significant changes demand the installation of additional on-site reflectors.

However, there are drawbacks:

- ✓ Laser triangulation systems can take a lot of time to design and install. The work starts in computer aided design (CAD) software, which a technician typically uses to design the layout of reflectors across the site. This is followed by the installation of the reflectors themselves, followed by several days' work by a qualified surveyor to accurately measure the reflectors' positions and create the map the AGVs will use.
- ✓ The LiDAR laser scanners used for vehicle positioning (also called localization) are typically mounted high on the vehicle in order to identify reflectors and avoid humans or moveable objects that might hide these references. This additional component increases vehicle cost and effectively constrains a vehicle maker to certain types of vehicle as, for example, there is nowhere to place such an extra sensor on an automated guided cart (AGC) or 'mouse' AGV.

1.2.4 VISION GUIDANCE:

Vision guidance is similar to how humans look at the world. It uses cameras in place of our eyes to recognize features in the environment. These are then compared to a 3D map, allowing the vehicle to calculate its position and navigate.



Fig 1.5 Vision guided AGV

The installation of vision guidance AGVs is fairly simple and quick, as vision guidance does not require any alterations to a site's existing infrastructure: the user simply drives the vehicle along the route, continuously collecting images all the while, in order to learn the path and construct the 3D map.

The time and costs of maintaining a vision-based system are also low, as changing routes just requires an operator to re-teach the vehicle, driving it manually around the new routes required.

However, this technology suffers from the usual drawbacks associated with vision-based systems:

- ✓ Changes in lighting conditions, such as glare or darkness, can impact the system's accuracy.
- ✓ Camera and image resolution can have similar effects, as can changes in the environment.

1.2.5 LiDAR FEATURE NAVIGATION:

LiDAR (Light Detection and Ranging) feature navigation (sometimes called natural or SLAM navigation) uses laser scanners, such as a vehicle's built-in safety scanners. It compares the current 2D scanned image to a map, created previously, to calculate the vehicle's position.

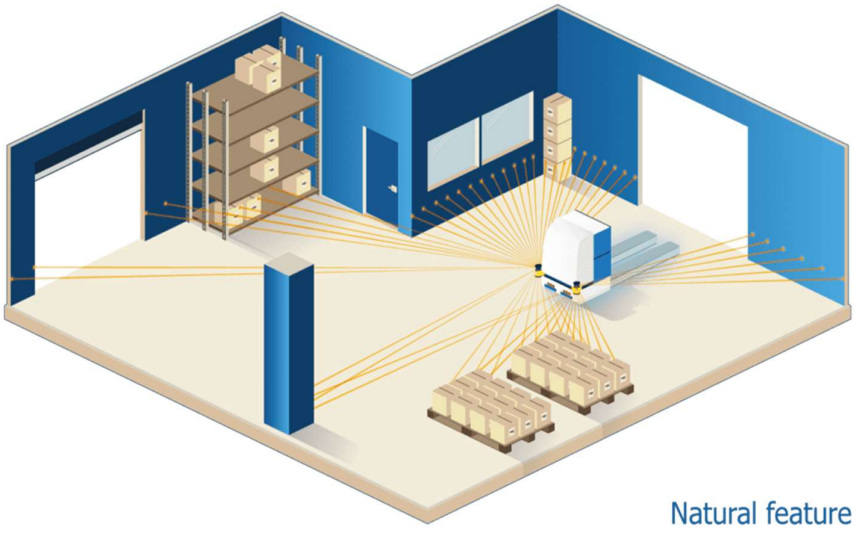


Fig 1.6 Navigation using LiDAR in AGV

Natural feature navigation technology works in one of two ways. These approaches are:

- ✓ Scan matching – scanner data is compared to, and constantly updates, a reference map.
- ✓ Feature matching – data is compared to permanent 'features' in the environment, such as walls and columns, without overwriting the reference map.

With scan matching, the raw laser data gathered is compared to the original map and this map is updated if new objects are detected. This approach requires the scan to be of high quality in order to match correctly. The type of map used – grid-based with low cell resolution – can have an adverse effect on a vehicle's positioning accuracy.

By contrast, with feature matching, a relatively small number of permanent environmental features – such as walls, columns and permanently installed machinery – are used as references to localize the vehicle. This method is highly robust as it requires few references (i.e. 5% of the environment or less) to ensure a vehicle's accurate localization.

Plus, as the map never changes – unless required by a major infrastructure change such as moving walls – accuracy is retained over time.

1.3 TYPES OF AUTOMATED GUIDED VEHICLES:

1. **Automated Guided Cart:**



Fig 1.7 AGV Cart

2. **Forklift AGVs:**



Fig 1.8 Forklift type AGV

3. **Towing AGVs:**



Fig 1.9 Towing type AGV

4. **Unit Load Handlers:**

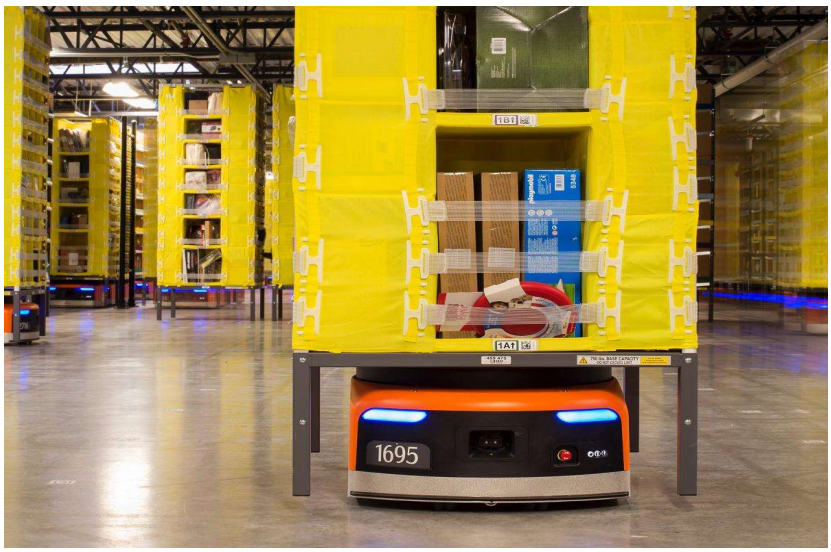


Fig 1.10 Unit Load Carrying AGV

5. **Heavy Burden Carriers:**



Fig 1.11 Heavy Load Carrying AGV

1.4 AGV STEERING:

- Differential speed control
- Steered wheel control
- Combination steering

Differential Speed Control:

This is the most common type of steering control used by AGVs. Differential speed control uses two independent drive wheels. Each drive wheel is driven at a different speed to turn. To go forward or backwards, the two drives are driven at the same speed. The simplest steering control option for AGVs, differential speed control doesn't require additional steering motors or mechanisms. It is commonly used for AGVs that operate in tight spaces or for those that operate near machines. It is not used for towing applications, as it can cause a trailer to jack knife while turning.

Steered Wheel Control:

This type of steering control is similar to the steering control in a car or truck. In steered wheel control, the drive wheel is the turning wheel. Steered wheel control is more precise than differential speed control and offers smoother turning. It is often used for towing applications and may also be operator-controlled.

Combination Steering:

This is a combination of differential speed control and steered wheel control. AGVs using combination steering have two independent steer/drive motors on diagonal corners of the AGV and swivelling castors on the other two corners. AGVs using combination steering can turn in any direction like a car and also drive in differential steering mode in any direction.

1.5 AGV TRAFFIC CONTROL:

A. ZONE CONTROL:

Simple to install and easy to expand, zone control is a commonly used traffic control method for AGVs. A wireless transmitter transmits signals in defined areas, and the AGV contains a sensor that receives the signal and transmits it back to the transmitter. If the area is clear, a "clear" signal is sent that allows the AGV to enter or pass through the area. If another AGV is in the area, a "stop" signal is sent that alerts other AGVs attempting to enter that the area is not clear. In this case, the waiting AGVs will stop and wait until the first AGV moves out of the zone and a "clear" signal is sent by the transmitter. Another way zone control can be used is by equipping each AGV with its own transmitter, allowing it to send a "do not enter" signal to other AGVs approaching the zone.

B. COLLISION AVOIDANCE:

AGVs using collision avoidance zone control are equipped with sensors that transmit a signal and wait for a reply to determine if an object is in front of it. These sensors may be sonic, which work like radar, or optical, which uses infrared sensors. Both work in a similar manner. Bumper sensors are another type of collision avoidance sensor. Many AGVs are equipped with bumper sensors as a fail-safe. Bumper sensors stop to avoid a collision when they sense physical contact.

C. COMBINATION CONTROL:

AGVs that use combination control are equipped with both collison control sensors and zone control sensors in order to offer more robust collision prevention in all situations. For instance, an AGV may use zone control as its primary traffic control system but also have collision avoidance sensors as a backup in case the zone control system malfunctions.

1.6 BENEFITS OF AGVs:

AGVs offer numerous benefits in warehousing and manufacturing.

A. Increase efficiency and productivity:

Because they operate autonomously, AGVs increase efficiency and productivity, and they're predictable and reliable for repetitive tasks. AGVs eliminate unnecessary walking and also eliminate the physical labour of transporting materials. They set the pace for workers, as well, keeping associates on-task. AGVs like collaborative mobile robots' guide associates through each task, reducing human error, which helps to improve order accuracy and minimize loss and misplaced products. By leveraging AI to optimize routes and prioritize work, collaborative mobile robots improve resource utilization.

B. Consistent costs:

AGVs are typically acquired on a per unit or per rental period cost basis, so there's less fluctuation in costs compared to human labour, which can fluctuate based on market conditions and demand.

C. Flexibility:

Some AGVs offer the flexibility of easily changing routes (compared to others which require re-routing guide wires or other infrastructure to adjust a vehicle's route). Automated guided vehicles are a scalable solution, as well, with the ability to add additional units based on demand.

D. Less space required:

Compared to other automation solutions, such as conveyor systems, AGVs require less space. Some AGVs are smaller compared to traditional warehouse equipment, such as forklifts, which allows for floor layouts with narrower aisles and better space utilization.

E. Improved safety:

Finally, AGVs are a safe automation solution for warehouses, distribution centres and manufacturing facilities. AGVs are equipped with sensors to avoid collisions. Advanced AGVs like AMRs have intelligent routing capabilities that enable them to plan the most efficient path through a warehouse or facility, reducing aisle congestion and preventing injuries.

Automated guided vehicles are uniquely designed to automate tasks for which humans are not necessary. They automate long walks in warehouses, set the pace for workers, and eliminate the physical rigors of a traditionally physically demanding job. Chuck by 6 River Systems is a flexible, scalable and cost-effective solution for warehouses and distribution centres offering innovative technology designed to tackle some of warehousing's biggest challenges.

CHAPTER 2
LITERATURE SURVEY

CHAPTER 2
LITERATURE SURVEY

1. AUTOMATED GUIDED VEHICLE:

Author: Trevor Skipp

High demands on manufacturers have left their shipping warehouses in havoc. Human error has a negative effect on safety, efficiency, and quality. These expenses are reduced with the introduction of an Automated Guided Vehicle, AGV. A driverless, intelligent forklift uses an optical path to quickly and safely traverse a warehouse. Its capabilities are enhanced by the ability to send and receive tasks through RF data communication.

2. DESIGN AND OPERATIONAL ISSUES IN AGV-SERVED MANUFACTURING SYSTEMS:

Author: Tharma Ganesharajah,
Nicholas G. Hall &
Chelliah Sriskandarajah

Automated Guided Vehicle (AGV) systems are already in widespread use and their importance for material handling is expected to grow rapidly. The advantages that such systems can offer include increased flexibility, better space utilization, improved factory floor safety, reduction in overall operating cost, and easier interface with other automated systems. This survey paper focuses on design and operational issues that arise in AGV systems. The objectives of the paper are to unify various lines of research related to AGVs and to suggest directions for future study. We consider problems arising in flowpath design, fleet sizing, job and vehicle scheduling, dispatching and conflict-free routing. Flowpath design problems address computationally intractable issues in the physical layout of a single loop and complex networks. Transportation and related models, waiting line analysis and simulation approaches are used to address fleet sizing questions. Scheduling issues focus on three flowpath layouts. In line layouts, the most important issues include finding an efficient job sequencing algorithm and identifying optimal AGV launch times. In loop layouts, issues such as joint scheduling of the job and AGV schedules, interface with a larger manufacturing system, dynamic job arrivals, and the location of the AGV parking area, are important. For complex network layouts, joint scheduling, heuristic dispatching rules, and conflict-free routing of AGVs, are considered. We

identify the inefficiencies that result from addressing these issues in isolation, suggesting the need for integration.

3. AGV SCHEDULING FOR AUTOMATED MATERIAL DISTRIBUTION:

Author: Naita Singh,
P. V. Sarngadharan &
Prabir K. Pal

We have studied the scheduling of automated guided vehicles (AGVs) for efficient and uniform material distribution from a truck-dock to machining units of the machine shop in an automotive manufacturing plant. The material distribution problem, being a much simpler special case of the more general material transfer problem, is easily amenable to analysis. We have assessed the number of AGVs required to meet the total material requirements of all the machining units in the shop. Proposing innovative dispatch rules, we have evaluated their performance in simulation by monitoring parameters reflecting efficiency and uniformity of material distribution, both for single AGV and multiple AGV case. In multiple AGV case, we introduce the notion of zones having comparable demands for AGV, and assign one AGV to each zone, so that each AGV can operate largely independently—sharing a minimum path with other AGVs using deadlock avoiding protocols. The results of simulation runs and their implications are discussed.

4. AUTOMATED GROUND VEHICLE (AGV) AND SENSOR TECHNOLOGIES:

Author: Liam Lynch

Automatic Ground Vehicles (AGVs), require guidance systems to navigate through their environment. The navigation system and sensors employed are dictated by the ambient conditions in that environment and the task to be completed by the AGV. This paper examines AGVs, the types of navigation systems available and the sensors employed by these systems.

CHAPTER 3
DESIGNING AN AGV

CHAPTER 3

DESIGNING AN AGV

3.1 INTRODUCTION:

The important components and its specification need to be chosen properly. The most important parts to be considered for robot design are Wheel Diameter of both Main drive wheel and Supporting wheel and their arrangement. Material on the wheel is also an important factor, because it distributes the load to the ground. Weight of Internal component and weight it can handle is also an input for the design process.

3.1.1 WHEEL ARRANGEMENT:

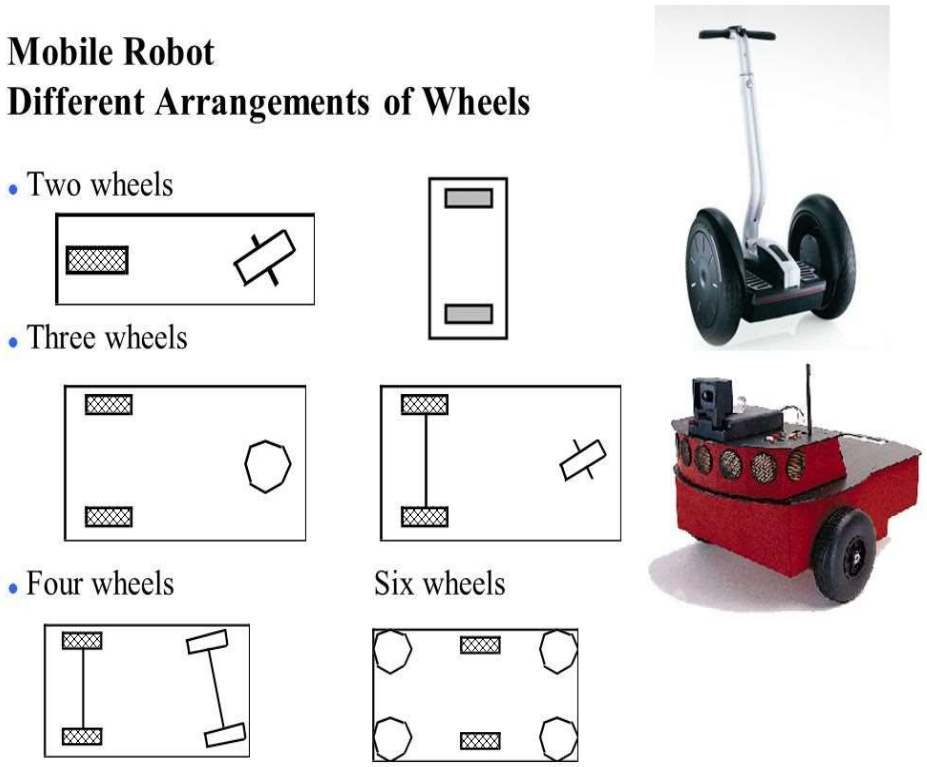


Fig 3.1 Different Wheel Arrangement

In this wheel arrangement we have two functions:

1. **Supporting Wheel**
2. **Direct Drive Wheel**

Supporting Wheel:

These wheels don't rotate on their own, they are rotate when a Direct drive wheel pushes the Vehicle. It just provides support to the vehicle to stabilize the load applied on the body of the AMR.

Direct Drive Wheel:

These wheels are in contact with the motor and gearbox and causes the vehicle motion. These are responsible for direction control and also for breaking/stopping. They are also called as Main Drive Wheels.

3.1.2 WHEEL SIZE AND MATERIAL:

Wheel sizes vary from design to design so they are not standard for all AMRs. For wheel material we use Polyurethane rubber , they are also called the Polyurethane Traction Wheel.

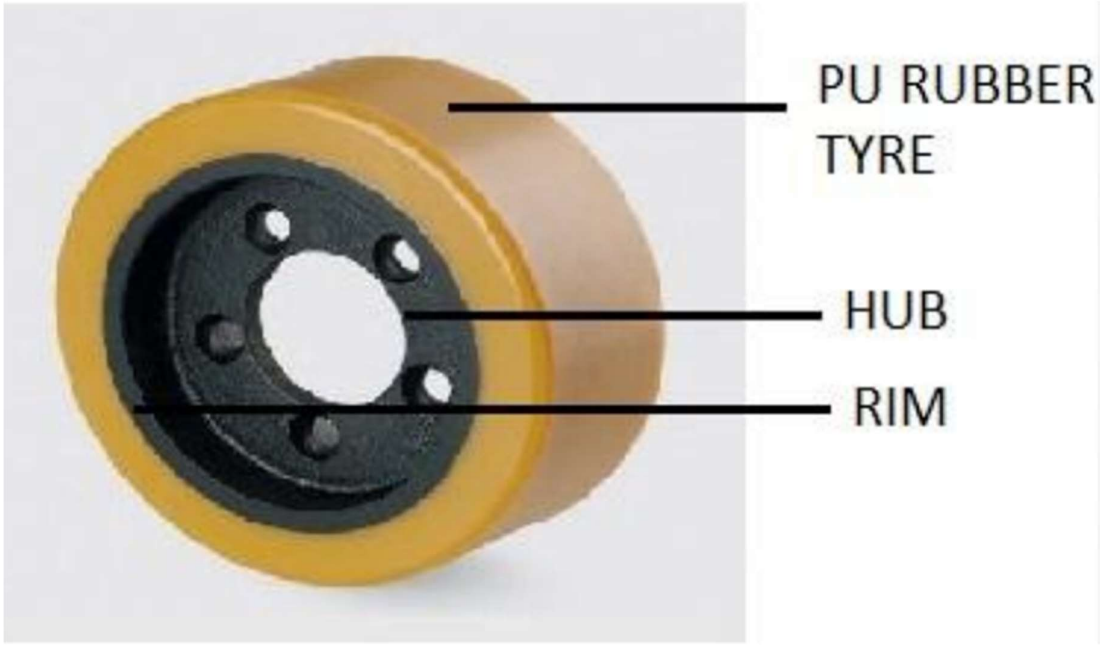


Fig 3.2 Polyurethane Wheel

Polyurethane wheels maintain their shape well and can operate efficiently at both high and low temperatures. Polyurethane as a tread material is non-marking and will not damage floors, making it the ideal choice for most industrial wheels and industrial castors.

Usability:

- ✓ They are found in different shape and size.

- ✓ They can be Machined to Required Shape, and Size.
- ✓ Fixing the Wheel to the body can be clamped with a Motor and by Just fixing the frame to Wheel's center using Ball Bearing with a Shaft.
- ✓ It can used as Supporting Wheel or as a Drive Wheel based in their Design.

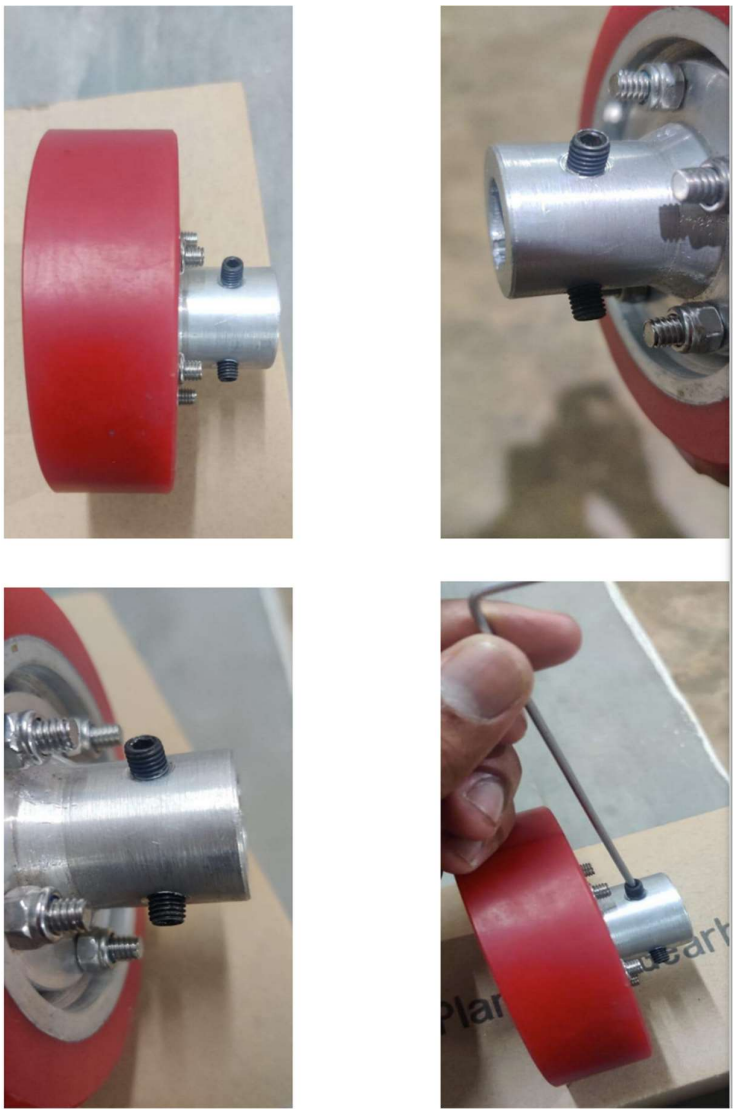


Fig 3.3 Reference Wheel used in our AMR

Advantages:

1. Noise Reduction:
    - ✓ The polyurethane acts as a shock absorber and helps to cushion the ride, absorbing bumps from uneven terrain. The noise levels can be reduced dramatically when using polyurethane instead of steel or cast-iron wheels.

2. Load Bearing Capability:
    - ✓ The material has higher tensile strength allowing it to carry heavier loads.
    - ✓ Since polyurethane is produced chemically, a number of variations are possible. The density and physical characteristics of the material can be modified according to the end use.
    - ✓ Polyurethane has excellent elastic memory. The flexibility offered by the material allows it to be used in several applications where plastic will fall short.
3. Floor Protection:
    - ✓ It reduces concentration of load and stress on the floor and keeps the floor in the same shape for longer period of time
4. Better grip, traction and durability:
    - ✓ Polyurethane is resistant to cuts and tears making it extremely durable.
    - ✓ In the same conditions as other materials, polyurethane wears at a slower rate making it more long lasting.
5. Resistance to Corrosion:
    - ✓ It is resistant to sunlight, oxygen, ozone, and several other environmental factors.
    - ✓ It is resistant to damage caused by abrasive chemicals.

3.2 PARTS AND SOFTWARE USED:

Our AMR works using Vision Based System (i.e. using LiDAR, Cameras, Distance Sensors) . Vision Based System is the latest guidance technology which operates without continuously defined the pathway.

Parts Needed:

1. Base Frame.
2. Wheel Arrangement.
3. Wheels.
4. Main Motor Drive Arrangement.
5. Side Supporting Pillars.
6. Top Rigid Frame.
7. Full Body Cover.
8. Other Hardware Components.

Software Used:

1. PTC Creo 7.0
2. Solidworks
3. Fusion 360

Fig 3.4 PTC Creo Logo

Fig 3.5 Fusion 360 Logo

Fig 3.6 Solidworks Logo

The structure that we need to build for the AMR is designed using anyone of the software, here we have used Solidworks to design the Whole Body. Here each and every part is created separately and assembled together. This is how the Basic Structure.

3.3 BASIC STRUCTURE:

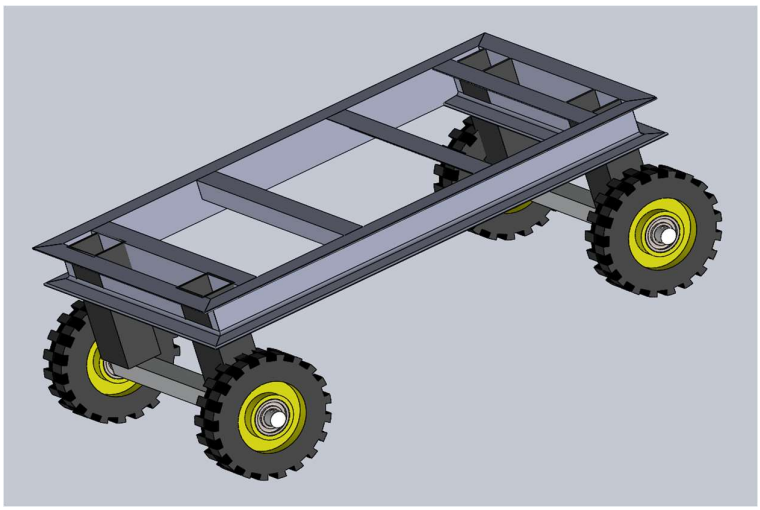


Fig 3.7 Isometric view of basic structure

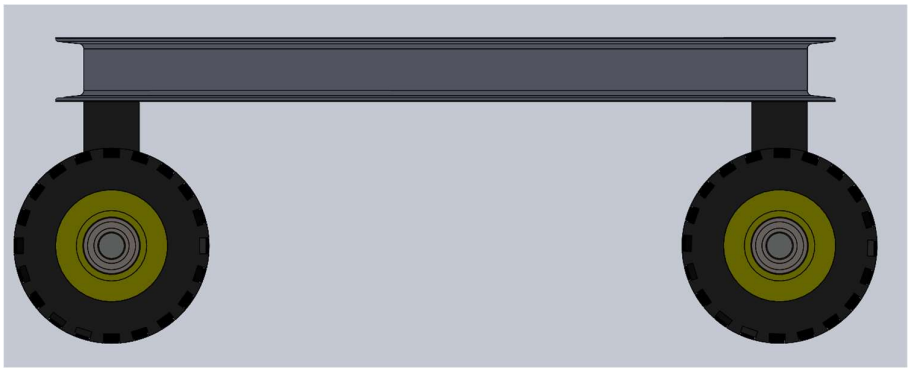


Fig 3.8 Right side view of basic structure



Fig 3.9 Front view of basic structure

From this basic structure we get an assumption of the final output. We can download the drawing files of some standard components and we can use it in our Assembly.

Other parts:

NEMA-34 STEPPER MOTOR:



Fig 3.10 Stepper Motor

PLANETARY GEARBOX:

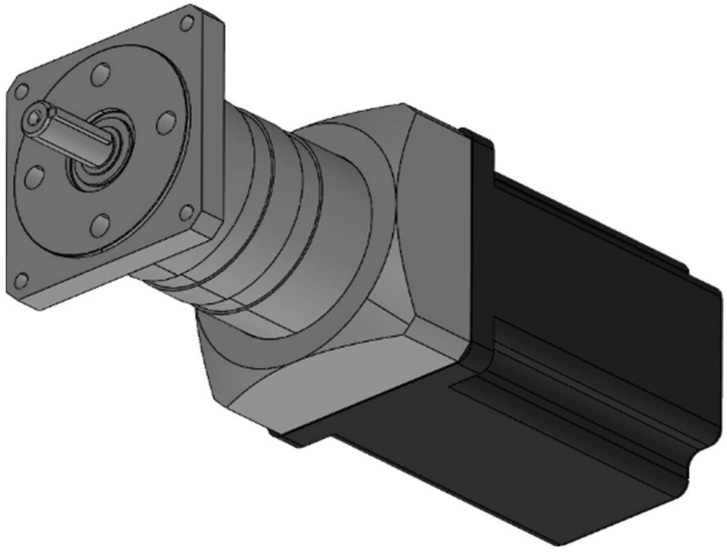


Fig 3.11 Planetary Gearbox

3.4 MAIN DRIVE AND SUPPORTING WHEEL ASSEMBLY:

MAIN DRIVE ASSEMBLY WITH WHEEL:

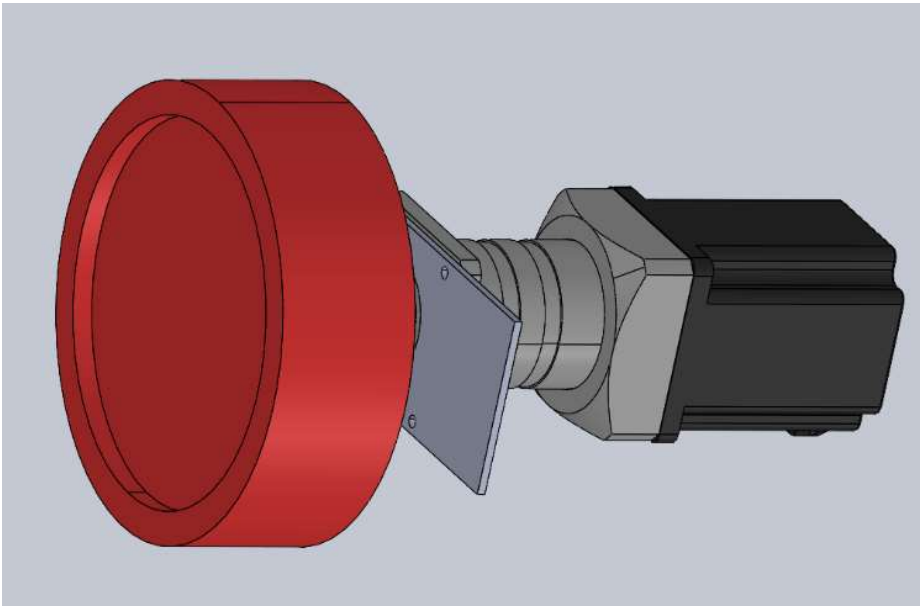


Fig 3.12 Main Drive

Here we can see the assembly of Main drive which consists of Stepper motor, planetary gear box, mounting plate and Wheel.

CASTER WHEEL ASSEMBLY:

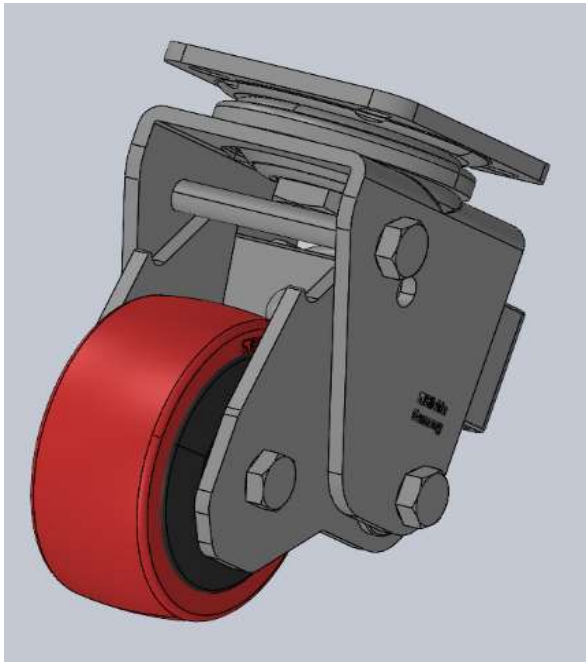


Fig 3.13 Front View of Supporting Wheel

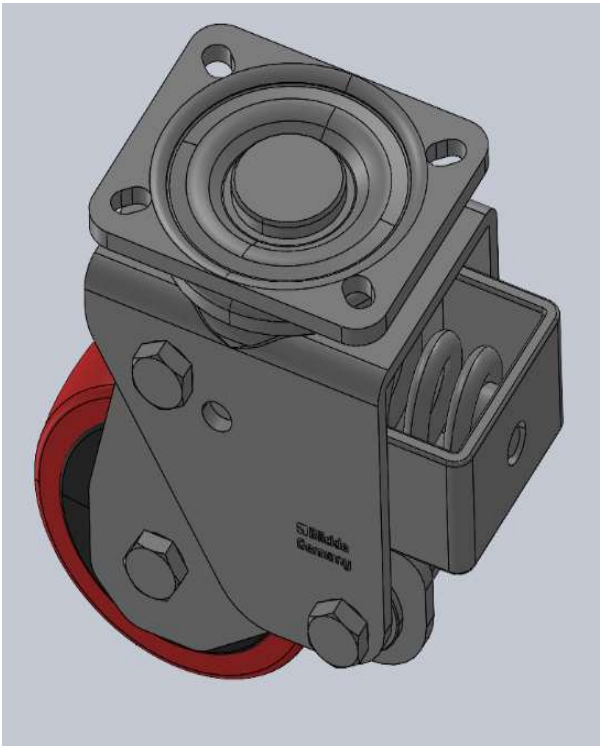


Fig 3.14 Mounting option of Supporting Wheel

MAIN DRIVE GUIDE:

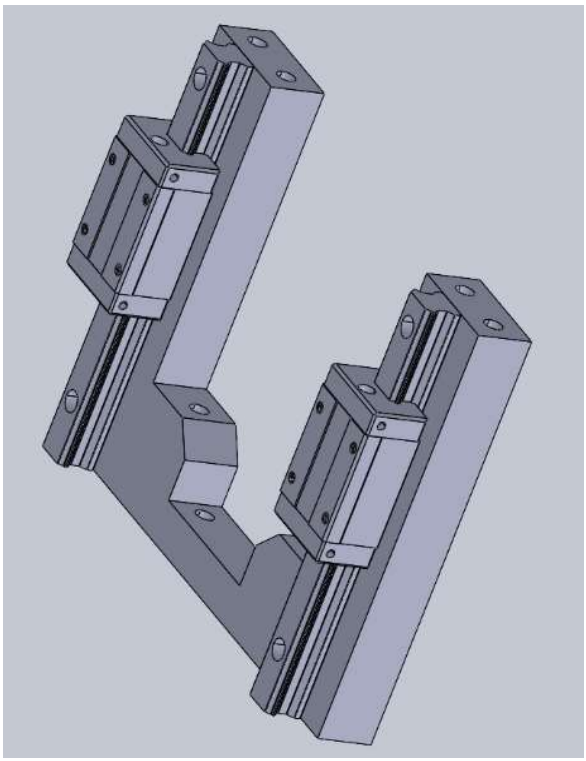


Fig 3.15 Linear Gear Guide

The linear gear guide is attached to the top frame with the Main Drive assembly, this helps foe height adjustment of the AMR, also acts a passive suspension

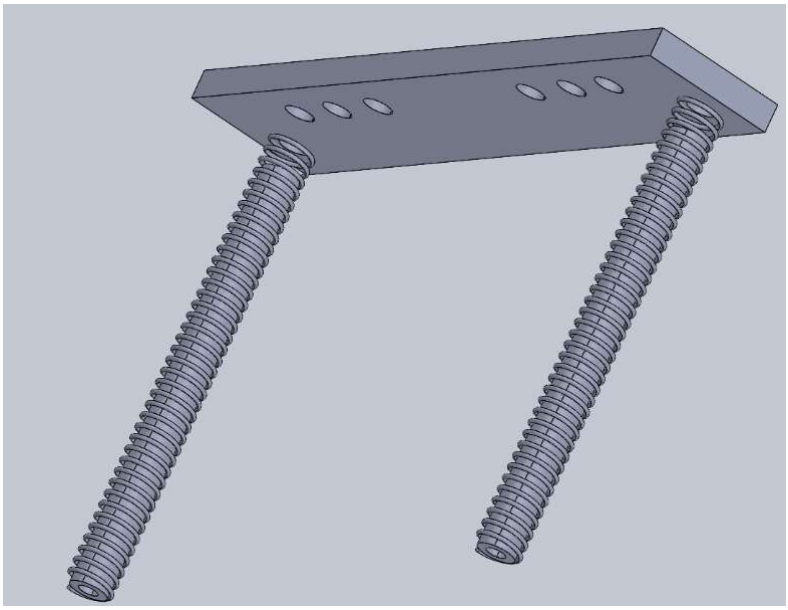


Fig 3.16 Suspension

3.5 MAIN FRAME :

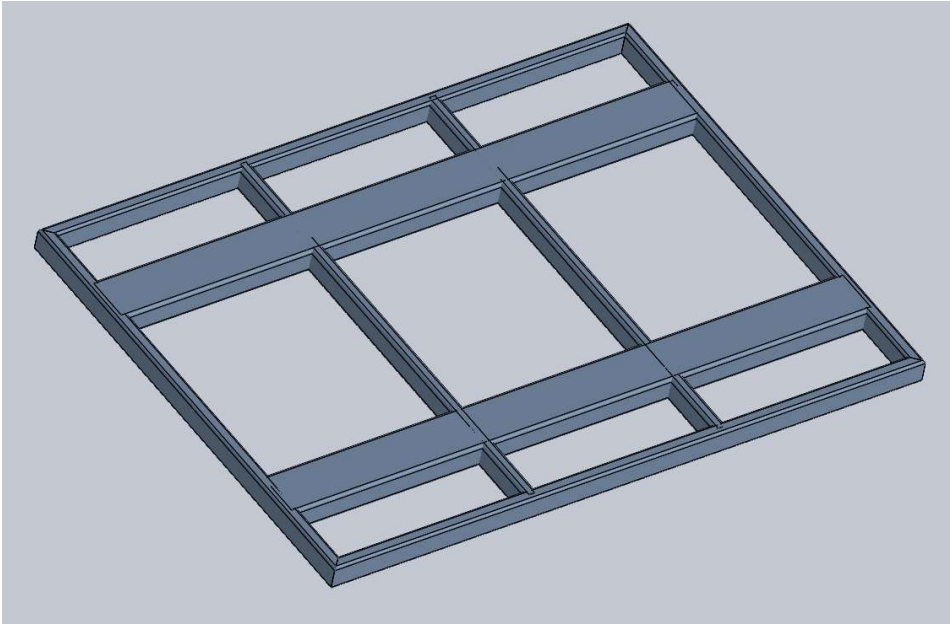


Fig 3.17 Top Rigid Frame

This is the heavy solid frame made of Mild Steel for handling heavy loads, this distributes the weight along the supporting beams and to the main drive suspension.

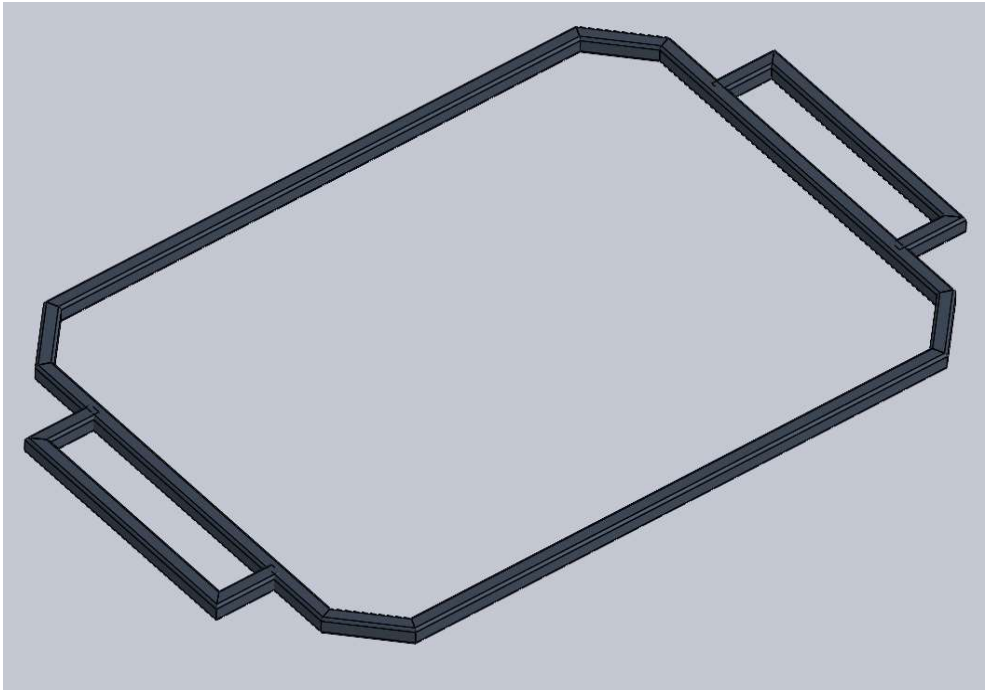


Fig 3.18 Bottom Frame

3.6 FINAL DESIGN:

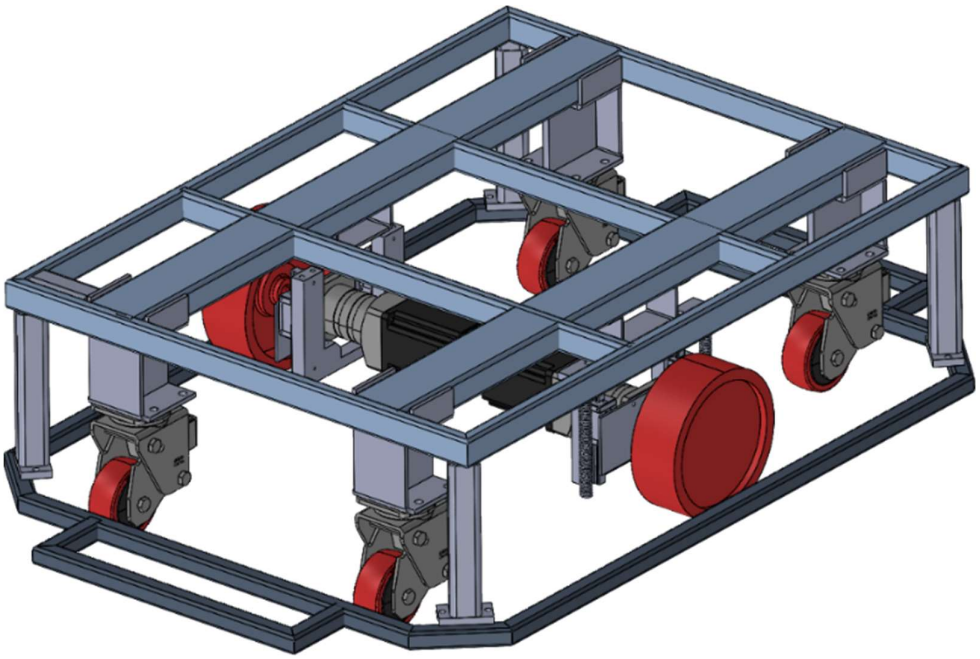


Fig 3.19 Isometric view of Final Design

The final design consists of the main frame, bottom frame, connected via supporting pillars like joints, Main Drive Assembly is attached to the suspension and the suspension is attached to the top frame, at last the supporting caster wheels are also attached to the top frame.

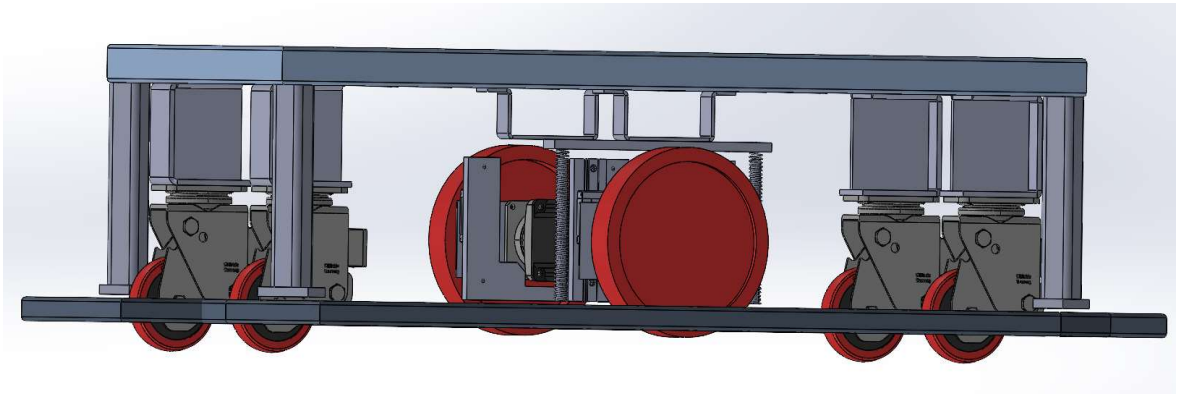


Fig 3.20 Side view of Final Design

CHAPTER 4
HARDWARE COMPONENTS

CHAPTER 4

HARDWARE COMPONENTS

4.1 NEMA-34 STEPPER MOTOR:

A Nema 34 stepper motor is a stepper motor with a 4.4 x 4.4-inch (86 x 86 mm) faceplate. Nema 34 high torque stepper motors deliver great value with no quality sacrifice. These motors are engineered to provide the highest possible torque but minimize vibration and audible noise. A wide range of motor windings and stack lengths are readily available, or the motors can be customized to meet your machine requirements. We have 8-wire motors that can be connected in all possible configurations: series, unipolar or parallel to maximize flexibility for your application. We can also customize the windings to perfectly match your voltage, current and maximum torque at operating speed.



Fig 4.1 Stepper Motor

Control:

Feature	Specification	
Operation Modes	Selectable	Profile position, Velocity, Profile velocity, Profile torque, Homing, Cyclic synchronous position
Display		Bi-colour LED
Software Tools	User Interface	Servo Studio, Windows-based
	Functions	Connection settings, Drive info, Power info, I/O configuration, Motion settings and tuning, Fault history/display
Rotary Units	Position Velocity Acc/Dec	Counts rpm/100 rpm/100/s

Table 4.1 Controls

Inputs/Outputs:

Feature	**Specification**	
1x Analog Input	Signal	Analog ±10 VDC differential
	Functions	User define
	Input Resolution	12 bits
	Input Impedance	94 kΩ
	Bandwidth (-3 db)	8 KHz
4x Digital Input **Exception:** **3x Digital Input on NEMA23 IP65**	Signal	Configurable opto-isolated. User defined compatibility with sinking or sourcing input. Exception: sinking input only on NEMA 17.
	Functions	Homing, limit switch, remote enable, start motion command for profiled position operation mode
	Voltage High Level Input	30 V
4x Digital Input **Exception:**	Min. High Level Input	11 V

3x Digital Input on NEMA23 IP65		
	Max. Low Level Input	5 V
	Input Resistance	2.2 kΩ
		Except: 24 kΩ NEMA17 IP20
	Max. Input Frequency	1 kHz
	Isolation Voltage	2500 Vrms
	Max. Input Current	According to max. voltage level, input current is not limited, drive limits the input current
	Propagation Delay Time	1 ms
2x Digital Output Exception: 1x Digital Output on NEMA23 IP65	Signal	Configurable open collector. User defined compatibility with opto-isolated sinking output or sourcing output.
	Functions	Motor speed set, Current, Motor speed set clear, Regen resistor control, Motion completed, in position, zero speed, Software position limit switch, Active, User selectable.
	Voltage	30 V
	Max. Current	500 mA
	Min. Load Resistance	60 Ω
	Output Voltage	0.25 V
	Min. Propagation Delay Time	1 ms (may be longer if load current is lower)

Table 4.2 Inputs/Outputs

4.2 PLANETARY GEARBOX:

The Sure Gear PGCN series is a great gearbox value for servo, stepper, and other motion control applications requiring a NEMA size input/output interface. It offers the best quality available for the price point.



Fig 4.2 Gearbox

Features

- ✓ Wide range of ratios (5, 10, 25, 50, and 100:1)
- ✓ Low backlash of 30 arc-min or less
- ✓ 20,000-hour service life
- ✓ Maintenance free; requires no additional lubrication
- ✓ NEMA sizes 17, 23, and 34
- ✓ Includes hardware for mounting to Sure Step stepper motors

Applications

- ✓ Material handling
- ✓ Pick and place
- ✓ Automation
- ✓ Packaging
- ✓ Other motion control applications requiring a NEMA input/output

4.3 YDLIDAR G2 LIDAR (12 M):

YDLIDAR G2 is a 360 degrees 2D LiDAR (hereinafter referred to as G2) developed by EAI team. Based on the principle of Triangulation, it is equipped with related optics, electricity, and algorithm design to achieve high-frequency and high-precision distance measurement. The mechanical structure rotates 360 degrees to continuously output the angle information as well as the point cloud data of the scanning environment while ranging.



Fig 4.3 LiDAR

Product Features:

- ✓ 360 degrees scan ranging
- ✓ Stable performance and high accuracy
- ✓ Wide measuring range
- ✓ Strong resistance to ambient light interference
- ✓ Industrial grade brushless motor drive, stable performance
- ✓ Laser power meets Class I laser safety standards
- ✓ 360 degrees omnidirectional scanning and 5-12Hz frequency
- ✓ Using optical and magnetic fusion technology to realize wireless communication and wireless power supply
- ✓ High-speed ranging, up to 5000Hz frequency

Applications:

- ✓ Robot navigation and obstacle avoidance
- ✓ Industrial automation
- ✓ Regional security
- ✓ Intelligent transportation
- ✓ Environmental scanning and 3D reconstruction
- ✓ Digital multimedia interaction
- ✓ Robot ROS teaching and research

Specifications:

Item	Min	Typical	Max	Unit	Remarks
Ranging frequency	/	5000	/	Hz	Ranging 5000 times per second
Motor frequency	5	7	12	Hz	Software control, Factory setting 7Hz
Ranging distance	0.12	/	16	m	80% Reflectivity
Field of view	/	0-360	/	Deg	/
Systematic error	/	2	/	cm	Range≤1m
Relative error	/	2.0%	/	/	1m<Range ≤8m
Luminous intensity range	0	/	1023	/	laser intensity
Tilt angle	0.25	1	1.75	Deg	Laser tilt angle
Angle resolution	0.36 (Frequency @5Hz)	0.504 (Frequency @7Hz)	0.864 (Frequency @12Hz)	Deg	/

Table 4.3 Specification of LiDAR

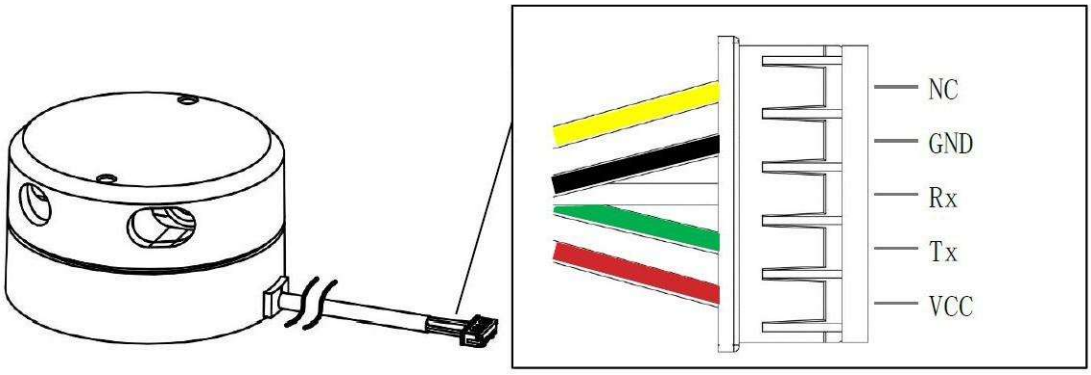


Fig 4.4 Pin Diagram

Pin Description:

Pin	Type	Description	Defults	Range	Remarks
VCC	Power supply	Positive	5V	4.8V-5.2V	/
Tx	Output	System serial port output	/	/	Data stream: LiDAR→Peripherals
Rx	Input	System serial port Input	/	/	Data stream: Peripherals→LiDAR
GND	Power supply	Negative	0V	0V	/
NC	Reserve	Reserved pin	/	/	/

Table 4.4 Pin Description

Working Mechanism:

The YDLIDAR G2(hereinafter referred to as G2) system has 4 working modes:

- ✓ **Idle mode:** When G2 is powered on, the default mode is idle mode. In idle mode, the G2's ranging unit does not work and the laser is not light.

- ✓ **Scan mode:** When G2 is in scanning mode, the ranging unit turns on the laser. When the G2 starts to work, it continuously samples the external environment and outputs it in real time after background processing.

- ✓ **Stop mode:** When G2 runs with an error, such as turning on the scanner, the laser is off, the motor does not rotate, etc. G2 will automatically turn off the distance measuring unit and feedback the error code.

- ✓ **Power-down protection mode:** In this mode, Lidar needs to continuously receive the scan commands with a less than 3 second's transmission interval so that it will keep working. If the Lidar does not receive a continuous scan command, the system will automatically stop. This mode is not enabled by default.

System Workflow:

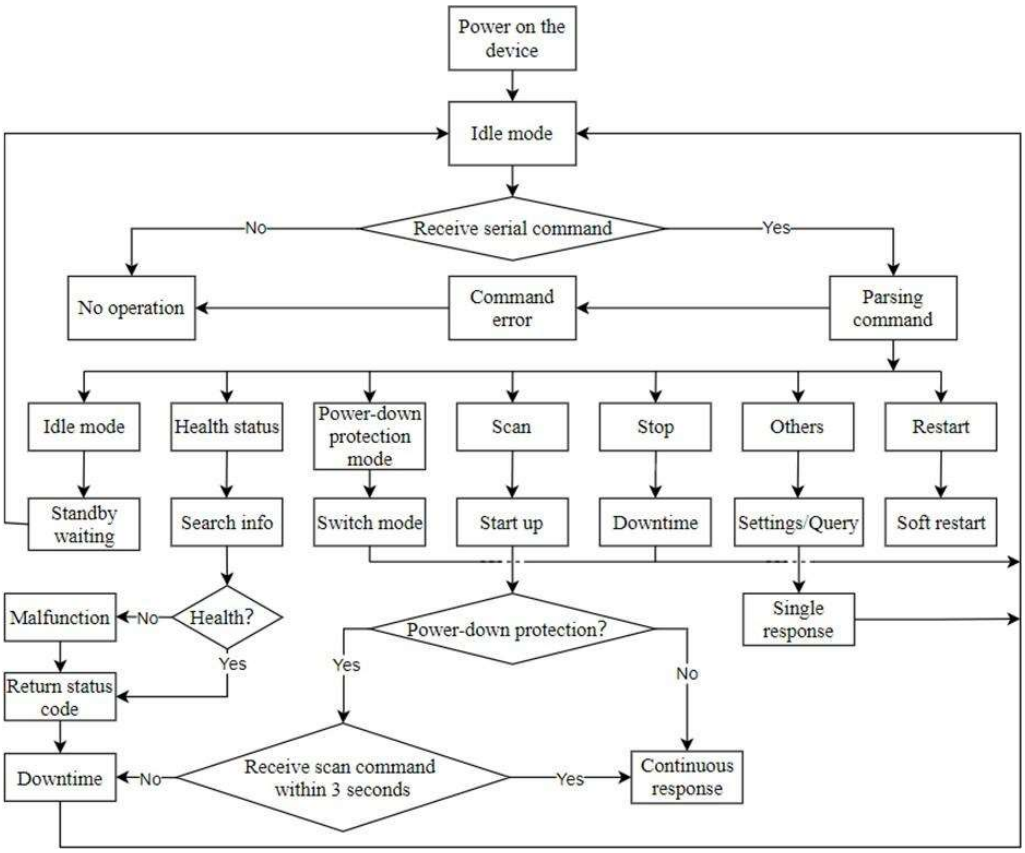


Fig 4.5 LiDAR Block Diagram

4.4 EMERGENCY STOP SWITCH-CE4T-10R-01:



Fig 4.6 Emergency Stop Switch

Technical Specification:

Ambient Air Temperature	Operation -25 ... +70 °C Storage -40 ... +85 °C
Colour	Red
Actuator Type:	Mushroom 40 mm
Pollution Degree:	3
Rated Insulation Voltage (Ui):	300 V
Rated Impulse Withstand Voltage:	6 kV
Rated Frequency (f):	50 / 60 Hz
Rated Operational Voltage:	24 ... 300 V
Rated Operational Current DC-13	(24 V) 0.3 A (125 V) 0.2 A
Rated Operational Current AC-15	(240 V) 1 A
Power Loss:	Maximum 2.5 W
Mechanical Durability	0.5 million
Connecting Capacity:	Flexible with Ferrule 1 x 0.5...1.5 mm² Flexible with Ferrule 2 x 0.5...1.5 mm² Flexible with Insulated Ferrule 1 x 0.5...1.5 mm² Flexible with Insulated Ferrule 2 x 0.5...1.5 mm² Flexible 1 x 0.5 ...1.5 mm² Flexible 2 x 0.5 ...1.5 mm² Solid 1 x 0.5...1.5 mm² Solid 2 x 0.5...1.5 mm²
Connection Type:	Screw Clamp

Table 4.5 Specification

4.5 DELTA DOP- B03S211 HUMAN MACHINE INTERFACE:

A human machine interface (HMI) is a platform which permits interaction between users and automation equipment. Delta's HMI products provide various communication ports for fast communication and convenient control of a diverse range of machines, systems and facilities.



Fig 4.7 Delta HMI

Specifications:

- ✓ 4.3-inch (480 x 272 pixels) TFT LCD 65536 colours
- ✓ 2 sets of COM port, supports RS-232 / RS-422 / RS-485
- ✓ For data transfer/download: RS-232, USB
- ✓ Complies with IP65 standard
- ✓ Supports horizontal / vertical display
- ✓ PC editing software, DOPSoft is compatible with operating systems: Windows XP, Windows Vista, Windows 7

Safety Precautions

Carefully note and observe the following safety precautions when receiving, inspecting, installing, operating, maintaining and troubleshooting. The following words, DANGER, WARNING and STOP are used to mark safety precautions when using the Delta's HMI product. Failure to observe these precautions may void the warranty.

Installation

- ✓ Comply with quick start for installation. Otherwise, it may cause equipment damage.

- ✓ Do not install the product in a location that is outside the stated specification for the HMI. Failure to observe this caution may result in electric shock, fire, or explosion.
- ✓ Do not install the product in a location where temperatures will exceed specification for the HMI. Failure to observe this caution may result in abnormal operation or damage the product.
- ✓ Please note that this equipment has obtained EMC registration for commercial use. In the event that it has been mistakenly sold or purchased, please exchange it for equipment certified for home use.
- ✓ Do not use this product as an alarm device for disaster early warning that may result in personal injury, equipment damage, or system emergency stop.

Maintenance and Inspection

- ✓ Do not touch any internal or exposed parts of the HMI as electrical shock may result.
- ✓ Do not remove operation panel while power is on. Otherwise, electrical shock may result.
- ✓ Wait at least 10 minutes after power has been removed before touching any HMI terminals or performing any wiring and/or inspection as an electrical charge may still remain in the HMI with hazardous voltages even after power has been removed.

4.6 SCHNEIDER ELECTRIC XB5AD21 SELECTOR SWITCH:



Fig 4.8 Selector Switch

Specification:

- ✓ Height: 42 mm
- ✓ Width: 30 mm
- ✓ Depth: 70 mm
- ✓ Terminals description ISO n°1: (13-14) NO
- ✓ Net weight: 0.043 kg
- ✓ Resistance to high pressure washer: 7000000 Pa at 55 °C, distance: 0.1 m
- ✓ Contacts usage: Standard contacts
- ✓ Positive opening: Without
- ✓ Torque value: 0.14 Nm. NO changing electrical state
- ✓ Mechanical durability: 1000000 cycles
- ✓ Tightening torque: 0.8…1.2 Nm. conforming to EN 60947-1
- ✓ Shape of screw head: Cross compatible with Philips no 1 screwdriver
    - o Cross compatible with posidrive No 1 screwdriver
    - o Slotted compatible with flat Ø 4 mm screwdriver
    - o Slotted compatible with flat Ø 5.5 mm screwdriver
- ✓ Contacts material: silver alloy (Ag/Ni)
- ✓ Short-circuit protection:10 A cartridge fuse type conforming to EN/IEC 60947-5-1
- ✓ Conventional free air thermal current: 10 A conforming to EN/IEC 60947-5-1
- ✓ Rated insulation voltage 600 V (pollution degree 3) conforming to EN 60947-1
- ✓ Rated impulse withstand voltage: 6 kV conforming to EN 60947-1

4.7 INTEL REALSENSE DEPTH CAMERA D450:



Fig 4.9 Intel Real sense Camera

Many different industries today use robots that can broadly be categorized as autonomous mobile robots, from delivery robots to those found in warehouses, factories, hospitals and more. An autonomous mobile robot is one which can navigate and move itself around spaces independently in service of its tasks. Once given appropriate tasks, the robot can act without the need for human guidance, finding its path and avoiding obstacles safely. AMRs require a solid understanding of their environment in real time to act – this is where Intel RealSense depth cameras come in. With high precision depth cameras that can operate in any environment, AMRs can get where they need to go, every time.

Block Diagram:

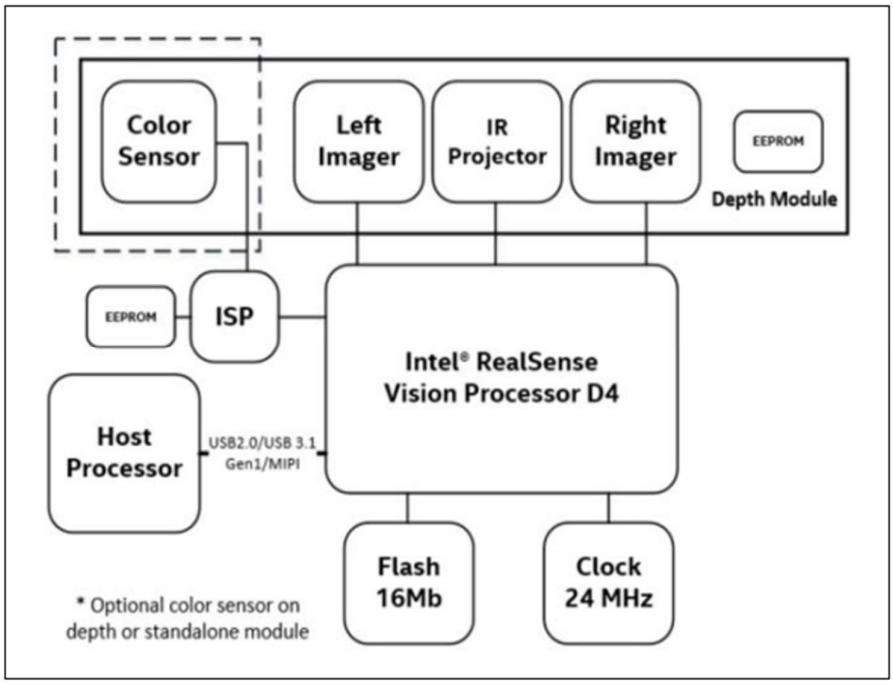


Fig 4.10 Camera Block Diagram

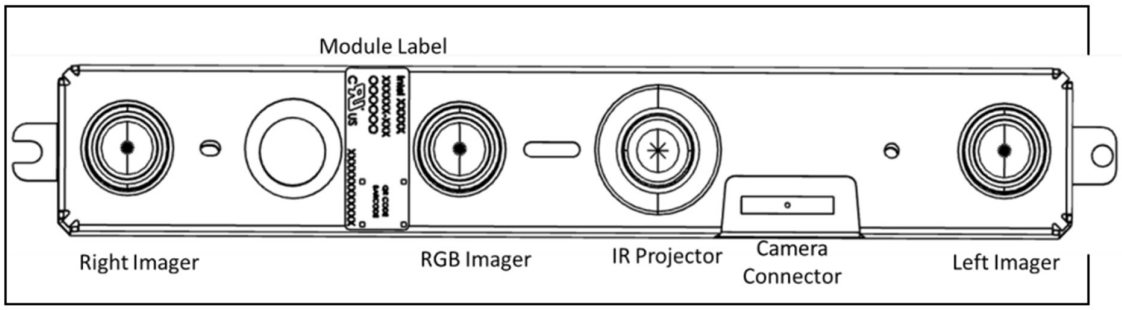
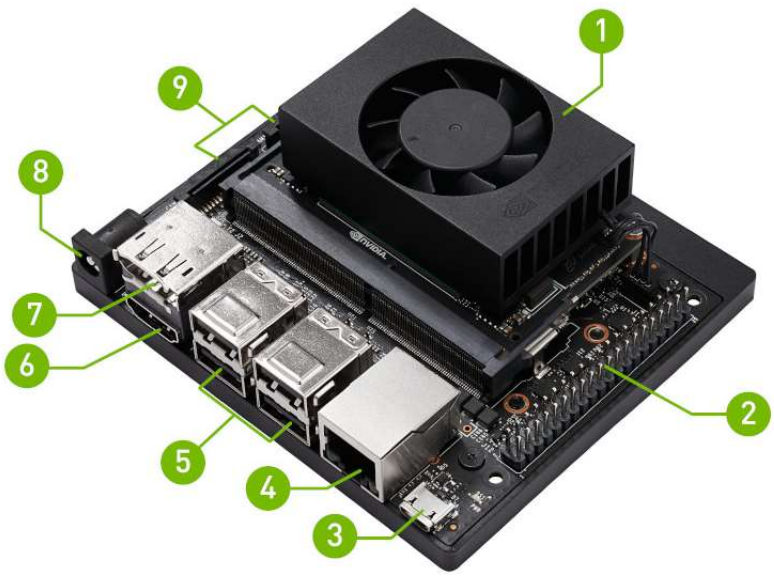


Fig 4.11 Stereo Depth Module

4.8 NVIDIA JETSON NANO:



1. microSD card slot for main storage
2. 40-pin expansion header
3. Micro-USB port
4. Gigabit Ethernet port
5. USB 3.1 Type A ports (x4)
6. HDMI output port
7. DisplayPort connector
8. DC Barrel jack for 19V power input
9. MIPI CSI camera connectors

Fig 4.12 NVIDIA Jetson Nano PORTS Description.

Processing Components:

- ✓ 6-core NVIDIA Carmel ARMv8.2 CPU @ 1400MHz (up to 1900MHz in dual-core mode)
- ✓ 384-core Volta GPU @ 1100MHz with 64 Tensor Cores
- ✓ Dual Deep Learning Accelerator (DLA) engines
- ✓ 7-way VLIW Vision Accelerator
- ✓ 8GB 128-bit LPDDR4x @ 1600MHz (51.2GB/s)
- ✓ 16GB eMMC 5.1
- ✓ (2x) 4Kp30 H.264/H.265 video encoder
- ✓ (2x) 4Kp60 H.264/H.265 video decoder

I/O Interfaces & Ports:

- ✓ (14x) MIPI CSI-2 lanes, 3x4 or 6x2 cameras up to 6 active sensor streams and 36 virtual camera channels.
- ✓ (2x) PCIe controllers | 1x4 (Gen 4), 1x1 (Gen 3)
- ✓ 1x4 (Gen 4) - Root Port and Endpoint
- ✓ 1x1 (Gen 3) - Root Port
- ✓ USB 4.1 (10Gbps) + (3x) USB 2.0
- ✓ (2x) DP 1.4 / eDP 1.4 / HDMI 2.0 a/b @ 4Kp60
- ✓ 10/100/1000 BASE-T Ethernet
- ✓ (3x) UART, (2x) SPI, (2x) I2C, I2S, CAN, GPIOs

Form-Factor:

- ✓ 260-pin SO-DIMM edge connector (backwards compatible with Jetson Nano module pin-out)
- ✓ dimensions: 70x45mm
- ✓ -25C to 90C operating temperature (Tj)
- ✓ 5VDC input power
- ✓ 10/15W power modes

Ports & Interfaces:

- ✓ (4x) USB 4.1 A (Host)
- ✓ USB 2.0 Micro B (Device)
- ✓ (2x) MIPI CSI-2 x2 (15-position Camera Flex Connector)
- ✓ HDMI 2.0
- ✓ DisplayPort 1.4
- ✓ Gigabit Ethernet (RJ45)
- ✓ M.2 Key-E with PCIe x1 (includes 2x2 802.11ac + BT 5.0 RTL8211 module)
- ✓ M.2 Key-M NVMe with PCIe x4
- ✓ MicroSD card slot
- ✓ (2x) I2C, (2x) SPI, UART, I2S, GPIOs

GPU	NVIDIA Volta architecture with 384 NVIDIA CUDA® cores and 48 Tensor cores
CPU	6-core NVIDIA Carmel ARM®v8.2 64-bit CPU 6 MB L2 + 4 MB L3
DL Accelerator	2x NVDLA Engines
Vision Accelerator	7-Way VLIW Vision Processor
Memory	8 GB 128-bit LPDDR4x @ 51.2GB/s
Storage	microSD (not included)
Video Encode	2x 4K @ 30 | 6x 1080p @ 60 | 14x 1080p @ 30 (H.265/H.264)
Video Decode	2x 4K @ 60 | 4x 4K @ 30 | 12x 1080p @ 60 | 32x 1080p @ 30 (H.265) 2x 4K @ 30 | 6x 1080p @ 60 | 16x 1080p @ 30 (H.264)
Camera	2x MIPI CSI-2 DPHY lanes
Connectivity	Gigabit Ethernet, M.2 Key E (WiFi/BT included), M.2 Key M (NVMe)
Display	HDMI and display port
USB	4x USB 4.1, USB 2.0 Micro-B
Others	GPIO, I^2C, I^2S, SPI, UART
Mechanical	103 mm x 90.5 mm x 34.66 mm

Table 4.6 Features of NVIDIA Jetson Nano.



Fig 4.13 NVIDIA Jetson Nano Developer Board

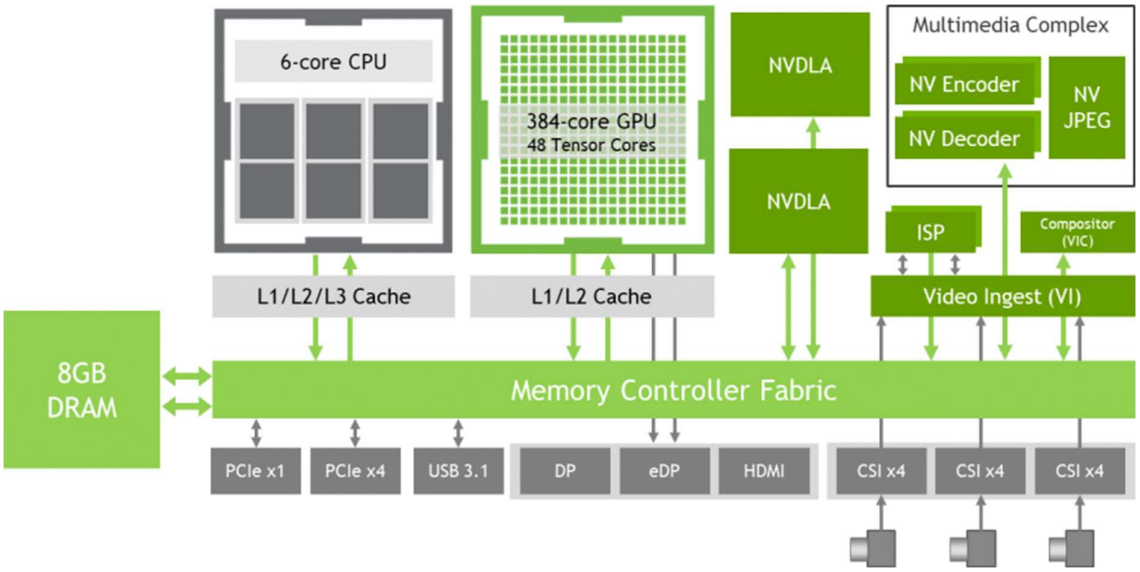


Fig 4.14 Architecture of Jetson development board

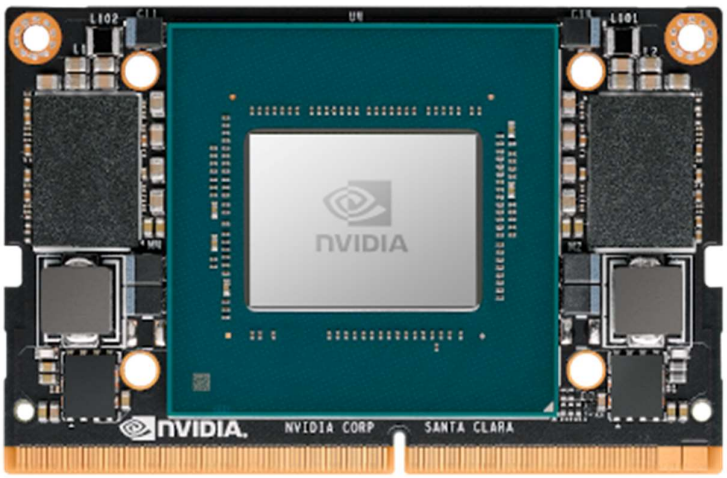


Fig 4.15 Jetson Processor board

4.9 BATTERY:

AGV series Lithium Iron Phosphate (LiFePO4) batteries have considerably greater energy density, making them the best choice for heavy equipment batteries, forklifts, AGVs, mobile robots, co-bots, ground support equipment, and warehouse equipment. Lithium iron phosphate presents an inherent safety performance and is more energy-efficient with a significantly longer cycle life than lead-acid.



Fig 4.16 Lithium-ion Battery for AGV

Features:

- Fast charging and low self-discharging
- Twice the energy provides twice the runtime
- Custom sizes for easy drop-in replacement
- No fire, no explore, sealed, maintenance-free design
- Flat voltage curve means no change in vehicle performance throughout the discharge
- Provides accurate State of Charge from the integrated battery management system
- Lithium phosphate batteries have 10 times the cycle life vs. lead-acid batteries

4.10 CHARGING DOCK:

Docking pads are charging systems used for charging batteries for Mobile Robots, AGV's. It eliminates the traditional way of charging batteries like manual plug-in charging. Customizable based on your requirement.

Can be integrated in the following areas like:

- AGVS
- Mobile/Service Robots
- Battery Charging/swapping in Automobile Production
- Warehouse and Packaging Facilities

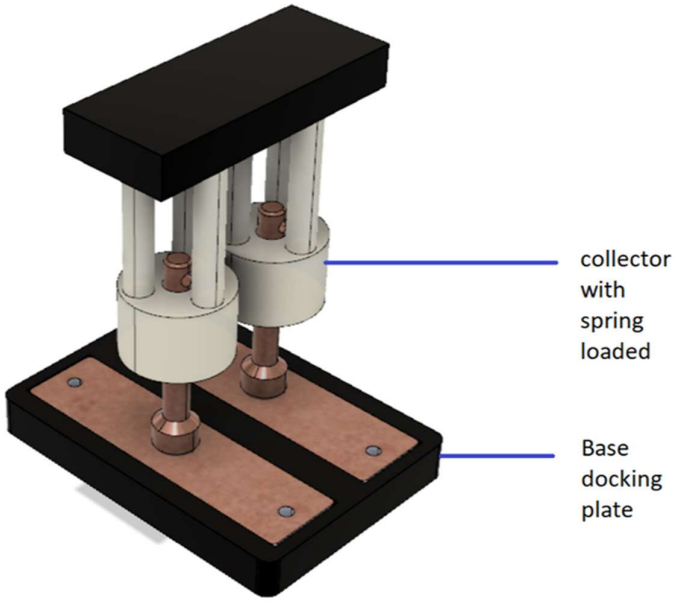


Fig 4.17 Auto Docking Pads For AGV



Fig 4.18 AGV is docked into the Docking Station for charging

CHAPTER 5
SOFTWARE ENVIRONMENT

CHAPTER 5

SOFTWARE ENVIRONMENT

5.1 ROBOT OPERATING SYSTEM:

Robot Operating System (ROS) is an open-source complex robot control software. The Robot Operating System (ROS) is a set of software libraries and tools that help you build robot applications. From drivers to state-of-the-art algorithms, and with powerful developer tools. Running sets of ROS-based processes are represented in a graph architecture where processing takes place in nodes that may receive, post and multiplex sensor data, control, state, planning, actuator, and other messages. ROS is also a global open-source community of engineers, developers and hobbyists who contribute to making robots better, more accessible and available to everyone.

5.2 TOOLS:

RVIZ:

RVIZ is a three-dimensional visualizer used to visualize robots, the environments they work in, and sensor data. It is a highly configurable tool, with many different types of visualizations and plugins.

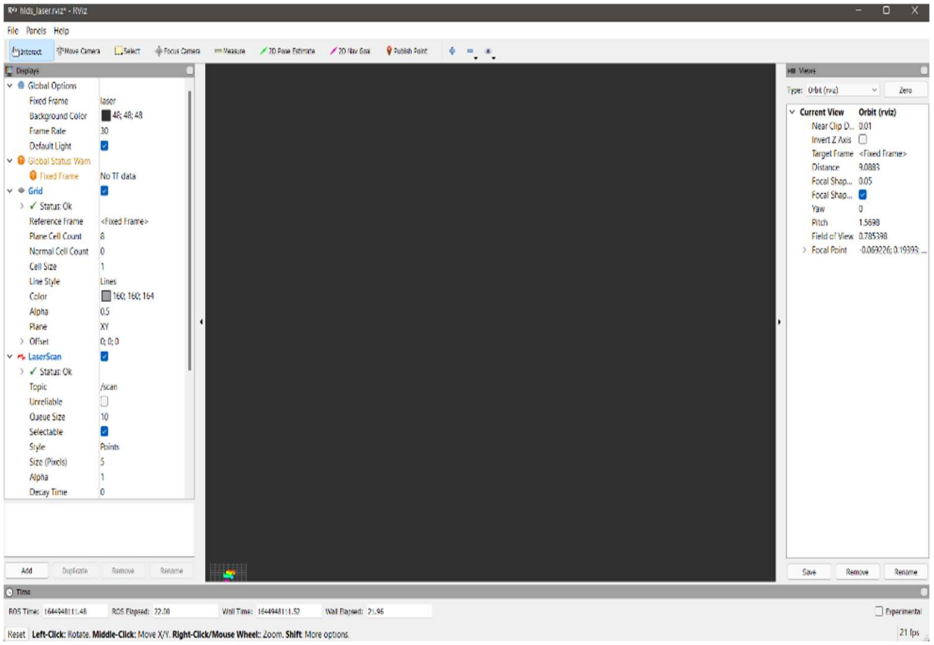


Fig 5.1 RVIZ Graphical User Interface

ROSBAG:

ROSBAG is a command line tool used to record and playback ROS message data. rosbag uses a file format called bags, which log ROS messages by listening to topics and recording messages as they come in. Playing messages back from a bag is largely the same as having the original nodes which produced the data in the ROS computation graph, making bags a useful tool for recording data to be used in later development. While rosbag is a command line only tool, rqt_bag provides a GUI interface to rosbag.

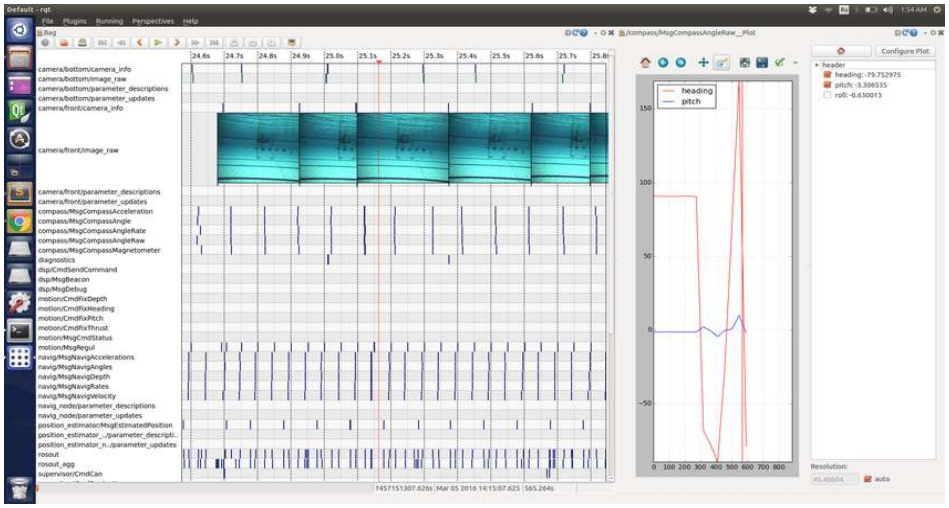


Fig 5.2 ROSBAG Sample Output

CATKIN:

CATKIN is the ROS build system, having replaced rosbuild as of ROS Groovy. catkin is based on CMake, and is similarly cross-platform, open-source, and language-independent.

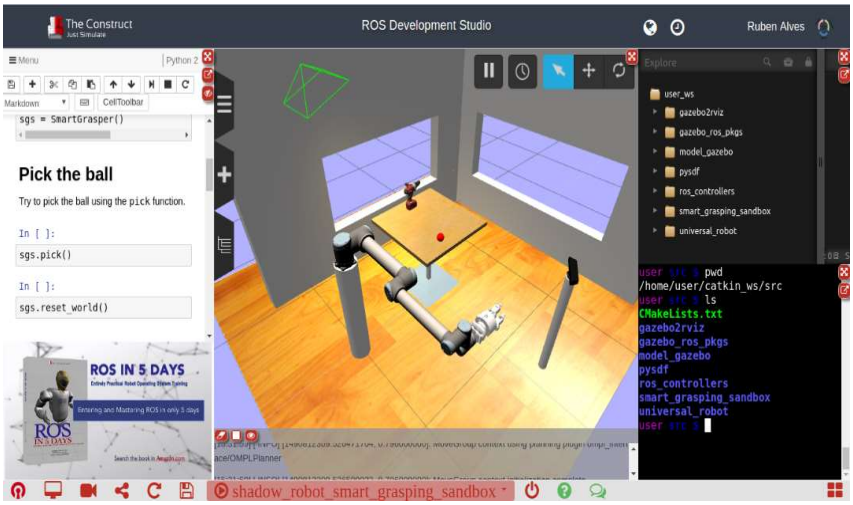


Fig 5.3 CATKIN ROS UI

ROSBASH:

The ROSBASH package provides a suite of tools which augment the functionality of the bash shell. These tools include rosls, roscd, and roscp, which replicate the functionalities of ls, cd, and cp respectively. The ROS versions of these tools allow users to use ros package names in place of the file path where the package is located. The package also adds tab-completion to most ROS utilities, and includes rosed, which edits a given file with the chosen default text editor, as well rosrun, which runs executables in ROS packages. rosbash supports the same functionalities for zsh and tcsh, to a lesser extent.

ROSLAUNCH:

ROSLAUNCH is a tool used to launch multiple ROS nodes both locally and remotely, as well as setting parameters on the ROS parameter server. roslaunch configuration files, which are written using XML can easily automate a complex startup and configuration process into a single command. roslaunch scripts can include other roslaunch scripts, launch nodes on specific machines, and even restart processes which die during execution.

5.3 VERSIONS AND RELEASES:

There are many versions but mostly we use only three:

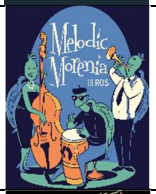
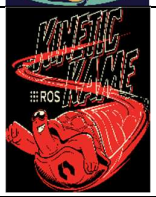
Noetic Ninjemys		**23 May 2020**
Melodic Morenia		**23 May 2018**
Kinetic Kame		**23 May 2016**

Table 5.1 Versions and Releases

5.4 SIRIUS:

It is a user interface which helps the operator to connect with the AGV. This is a completely customizable UI for customer needs and this is one of the customers requested UI

Home Page:

Home page is where one can Enable the Robot to accept Tasks and start monitoring the robot on the tasks.

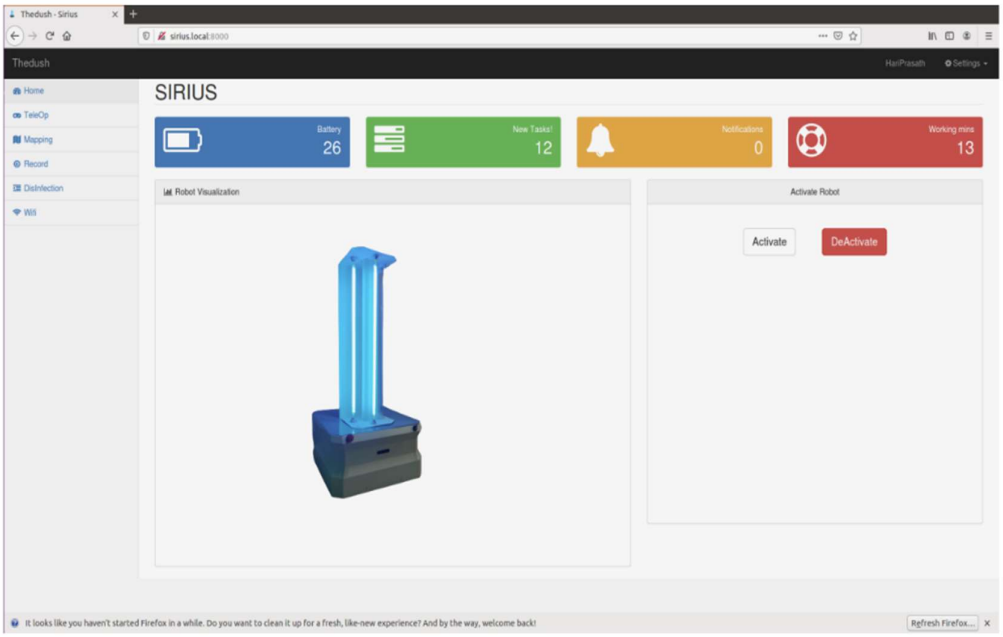


Fig 5.4 Sirius User Interface (Home page)

TeleOp Page:

TeleOp is the tab where the user can Activate the robot to accept Joystick commands if any Manual operation of the robot is needed. Once activated, the robot quits all the current tasks and starts accepting manual commands from the joystick presented in the UI.

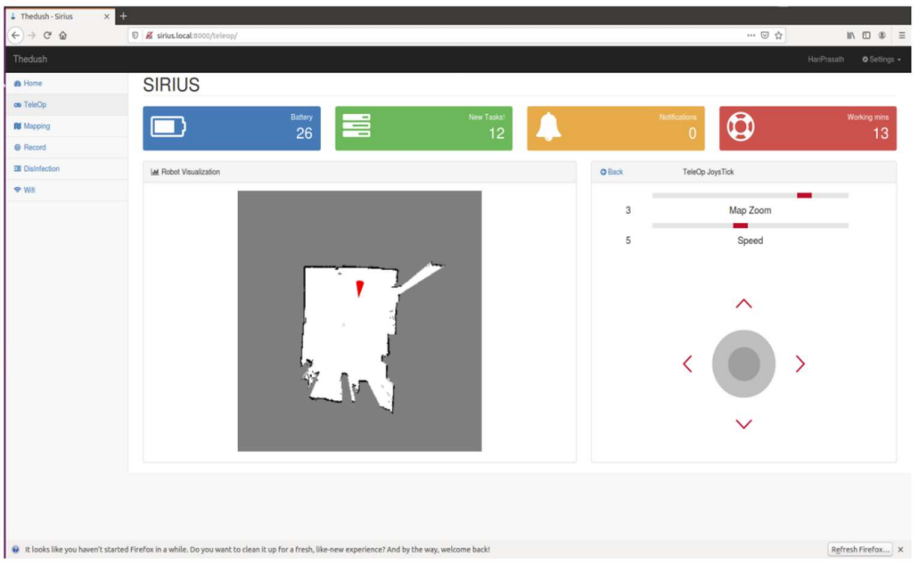


Fig 5.5 Teleoperation Page

Mapping Page:

Mapping is the tab where the initial mapping of the completely new Place is done. This can be easily done by activating the robot to start mapping the area and move along the path where you want to map. The UI also provides the user with the updated map simultaneously for easy mapping of the area. Once the map is saved, the user can then use the map to record the path in which they want the robot to move.

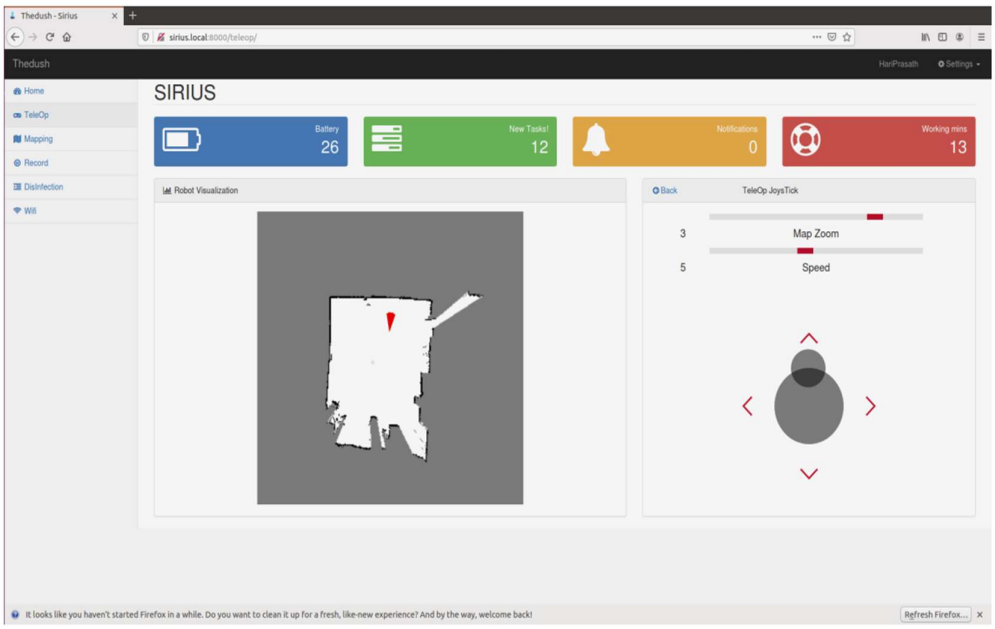


Fig 5.6 Mapping mode

Record:

The record page currently used to record the points in which the robot needs to move. Users can specify one or more points in which they want the robot to move. The robot then plans a shortest path between those points and reaches the destination.

Robots can maneuver around people and dynamic obstacles which drive independently in environments with people and obstacles. The robots scanners make sure they see people and obstacles and it acts in a natural way by slowing down, driving around the person or obstacle or stopping. This ensures a natural coexistence with humans.

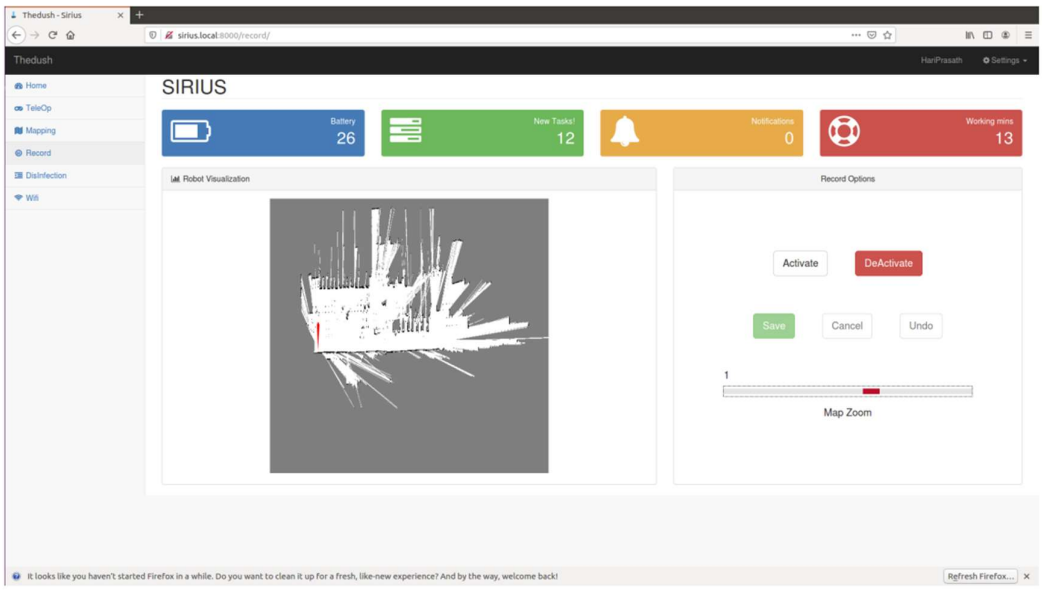


Fig 5.7 Recording the Map Generated

Disinfection:

The disinfection page is where the user can give a task to the robot to start the motion in the planned path and return to the home position. The robot can be programmed to automatically do ‘N’ number of cycles as the user seeks.

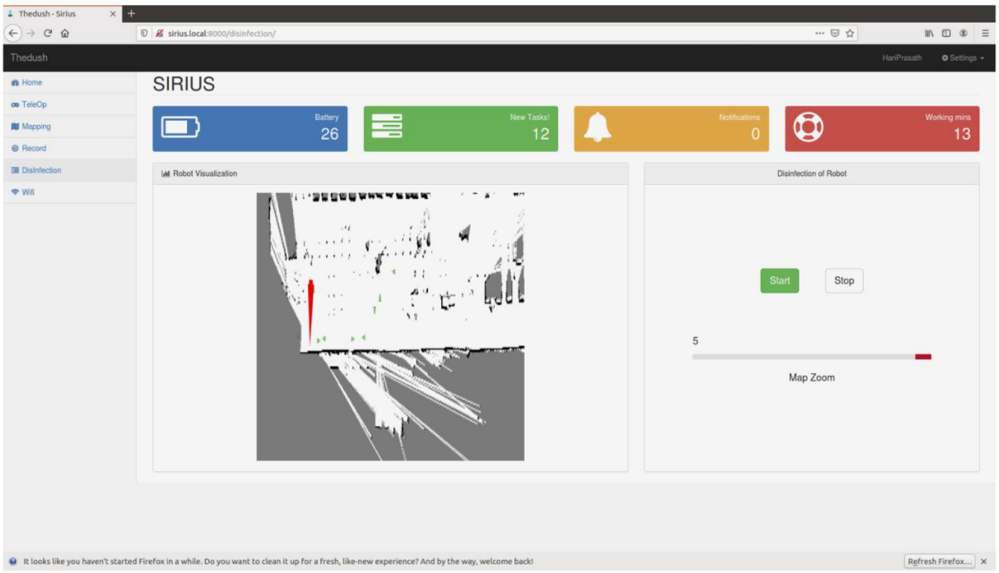


Fig 5.8 Disinfection Page

CHAPTER 6
SETUP

CHAPTER 6
SETUP

6.1 INSTALLING ROS – NOETIC:

Only Melodic and Noetic have support to run on windows using visual studio, so here we have chosen Noetic ROS package. Same is also installed in Ubuntu.

6.1.1 IN WINDOWS OPERATING SYSTEM:

There are some requirements to install ROS Noetic in WINDOWS Operating System:

1 64-bit Windows 10 or later versions.

2 PowerShell installed in the System path.

3 Turn-off Realtime Protech and any third-party antivirus software to not interrupt the installation process.

6.1.1.1 INSTALLING VISUAL STUDIO:

Building a ROS project for Windows requires Visual Studio 2019 and the Microsoft Windows 10 SDK.

1 Download Visual Studio 2019 installer.exe or later versions

2 Run the installer as administrator, and follow the instruction given in the prompt.

3 Vcpkg is used for managing dependent libraries. It requires that the English language pack to be installed.

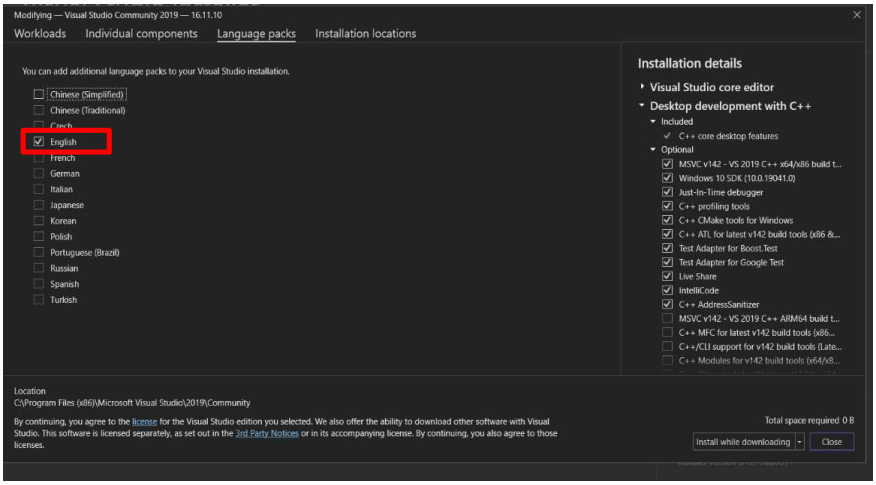


Fig 6.1 Installing Visual Studio

4 Include "Desktop development with C++" workload.

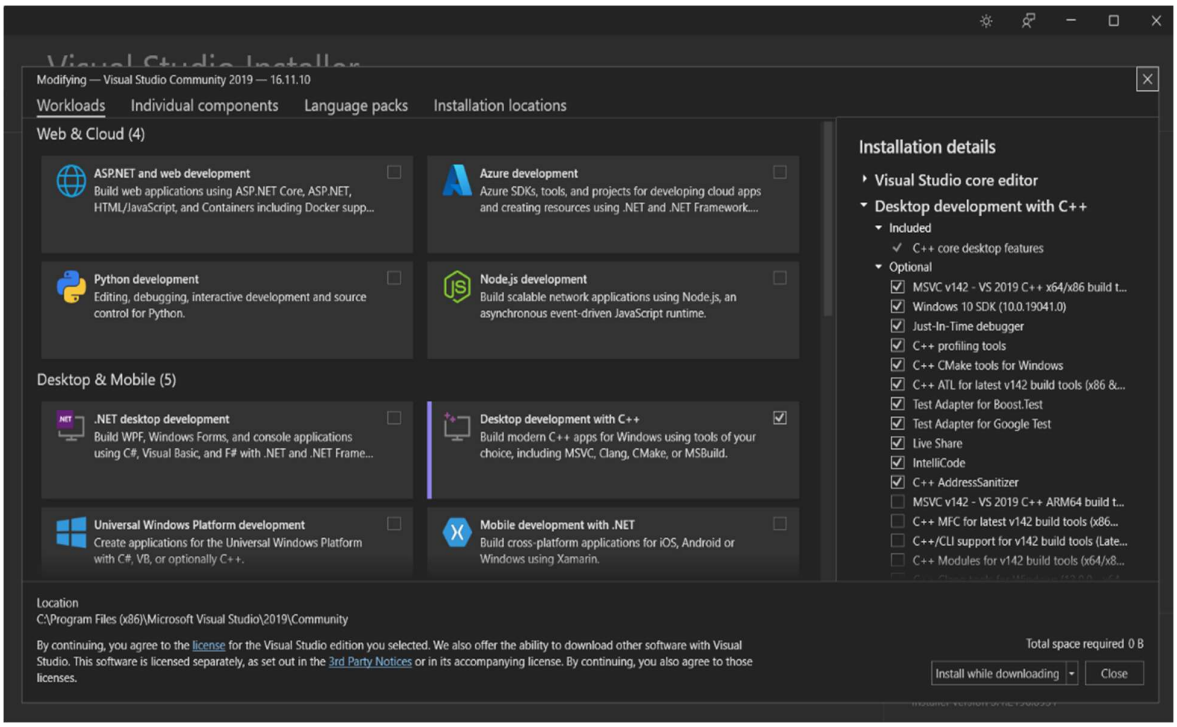


Fig 6.2 Packages Selection

5 In the Individual Components, select Windows 10 SDK (10.0.19041.0).

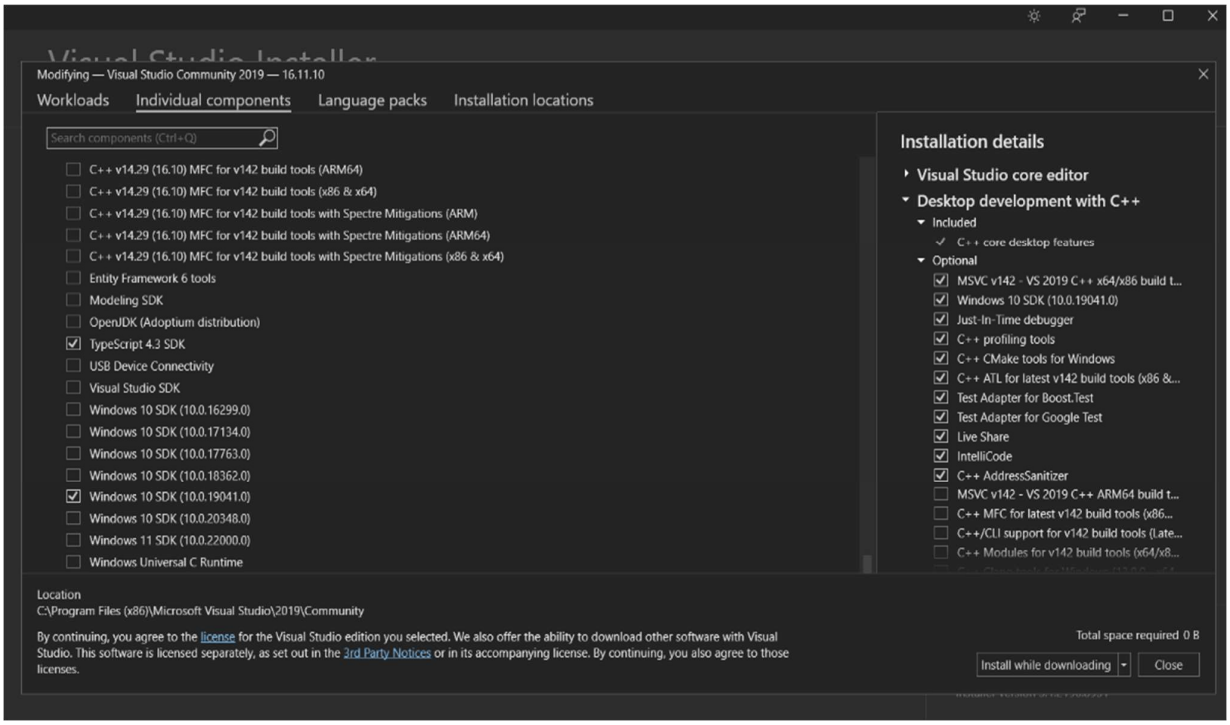


Fig 6.3 SDK Selection

6 Then Click on install, and this will take some time to download and install.

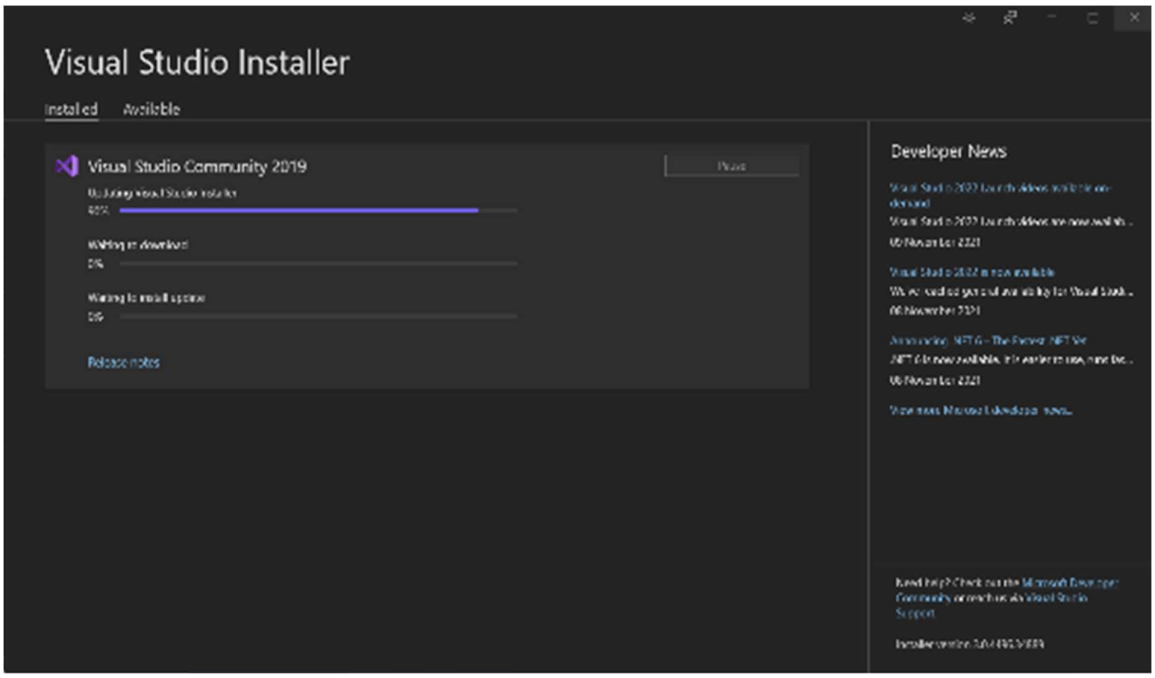


Fig 6.4 Installation Wizard

6.1.1.2 INSTALLING ROS:

1 In the Start Menu, find the ‘x64Native Tools Command Prompt for VS 2019 item

2 Right Click, select More then click on "Run as Administrator"

3 Copy and paste the following command line in the Prompt:

```
@"%SystemRoot%\System32\WindowsPowerShell\v1.0\powershell.exe" -NoProfile -InputFormat None -ExecutionPolicy Bypass -Command "iex ((New-Object System.Net.WebClient).DownloadString('https://chocolatey.org/install.ps1'))" && SET "PATH=%PATH%;%ALLUSERSPROFILE%\chocolatey\bin"
```

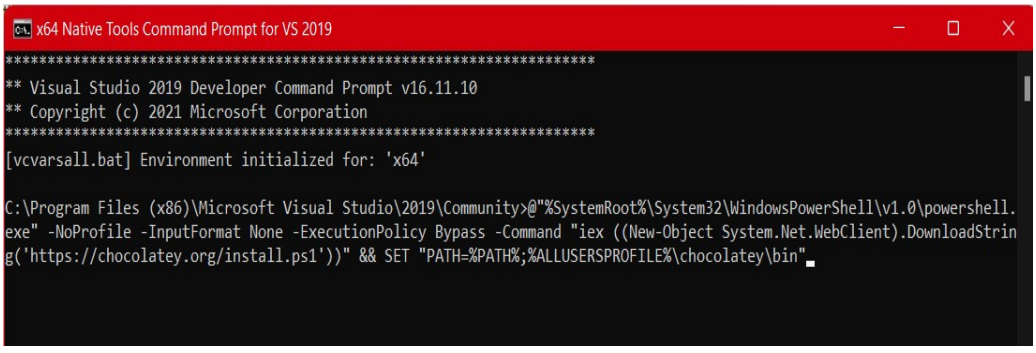


Fig 6.5 Installing ROS Via Command

4 Approve any prompts. Once it has completed, close the command prompt to complete the install.

Install GITHUB:

5 Reopen the Visual Studio Command Window as described above.

6 Please install Git using the command here, even if you have it installed as an application.

```
choco upgrade git -y
```

7 After Git installed, ensure Git is now available in the Visual Studio command window.

```
git --version
```

8 Close and Reopen the Visual Studio Command Window as described above.

ROS Last Known Good (LKG) Build Installation:

9 Open the Visual Studio Command Prompt as Administrator as described above.

```
mkdir c:\opt\chocolatey
set ChocolateyInstall=c:\opt\chocolatey
choco source add -n=ros-win -s="https://aka.ms/ros/public" --priority=1
choco upgrade ros-noetic-desktop_full -y --execution-timeout=0
```

Create a ROS Command Window shortcut:

10 Right click in a Windows Explorer folder, select New > Shortcut

11 In the shortcut path, copy the highlighted command line from the following options, depending on the Visual Studio install above:

```
C:\Windows\System32\cmd.exe /k "C:\Program Files (x86)\Microsoft Visual Studio\2019\Community\Common7\Tools\VsDevCmd.bat" -arch=amd64 -host_arch=amd64&& set ChocolateyInstall=c:\opt\chocolatey&& c:\opt\ros\noetic\x64\setup.bat
```

12 Name the shortcut "ROS"

13 Set that shortcut as Administrator:

a. Right Click on the shortcut and choose "Properties".

b. Select the Shortcut Tab if not already selected.

c. Press the Advanced button

d. Check the button "Run as Administrator".

e. Press OK on the Advanced properties dialog.

f. Press OK on the "ROS Properties" shortcut dialog.

6.1.2 IN LINUX BASED OPERATING SYSTEMS:

To run ROS in a Linux based operating system, in our project we have created a virtual machine for Ubuntu 20.04, and then installed ROS noetic in the same.

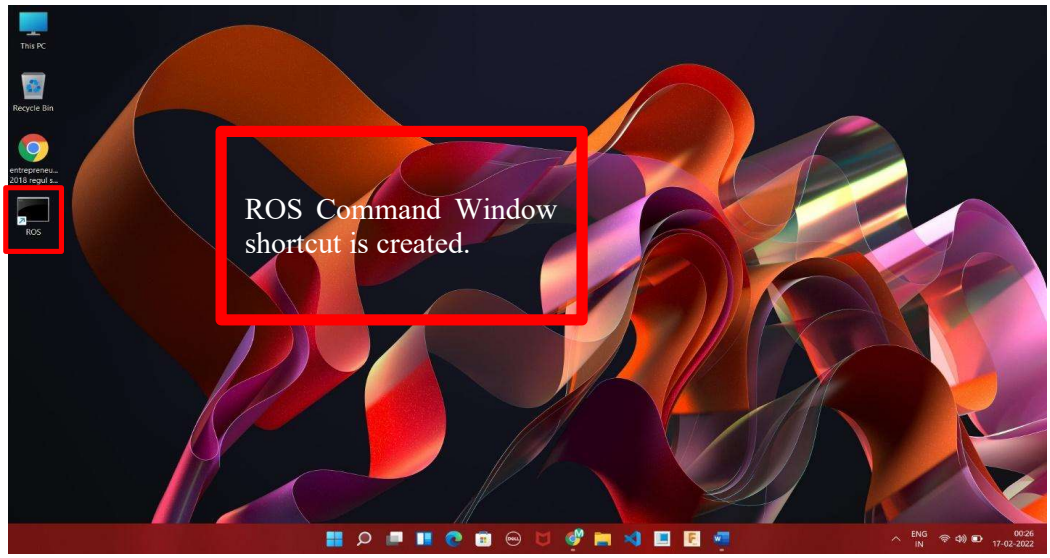


Fig 6.6 Shortcut Created

6.1.2.1 CREATING VIRTUAL MACHINE:

1. Open browser and search for Oracle VM VirtualBox and Download the Latest version, in our case version 6.1.32 was used.



Fig 6.7 Oracle Virtual Box Download Page

2. After Downloading, Install VirtualBox by Double Clicking on .exe file.



Fig 6.8 Virtual Box Installation Wizard

3. Click on Next and follow the Prompt. Click on Install to Begin Installation.

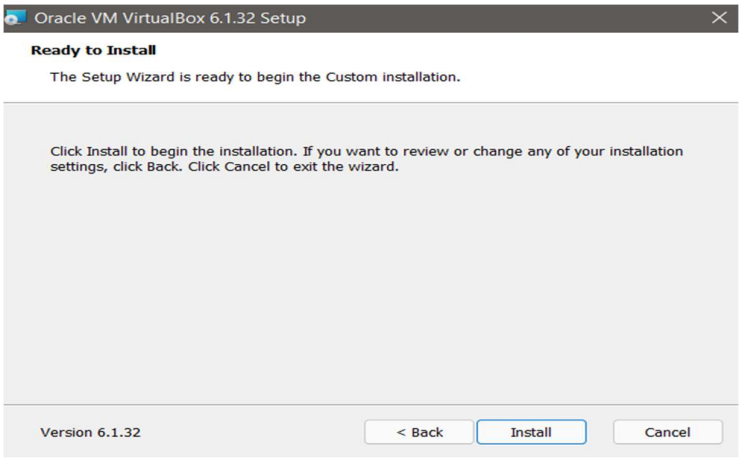


Fig 6.9 Virtual Box Setup

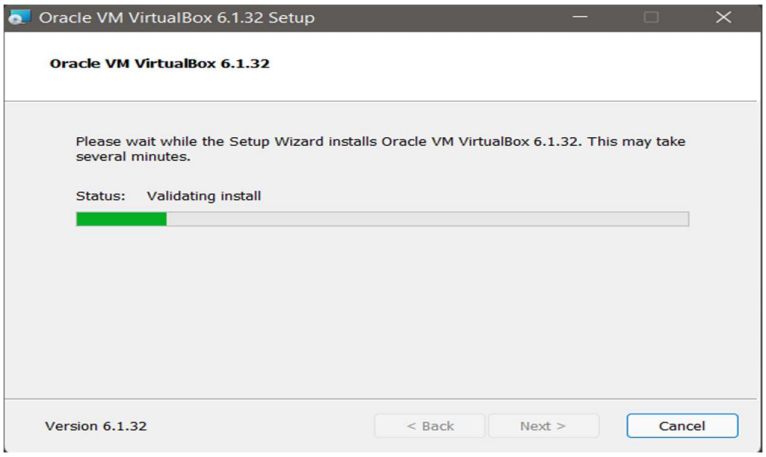


Fig 6.10 Installing

4. After Installation is Complete click on Finish.



Fig 6.11 Start Oracle VM VirtualBox

New Virtual Machine:

5. Click on New to create a Virtual Machine.

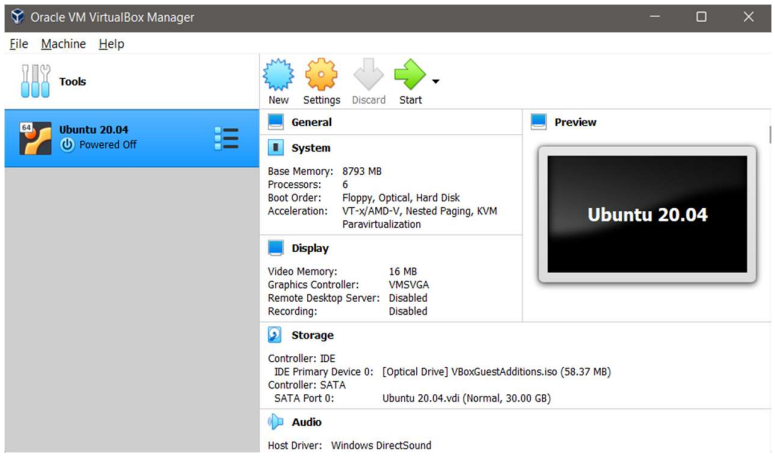


Fig 6.12 Virtual Machine User Interface

6. After installing open it, Click on New.

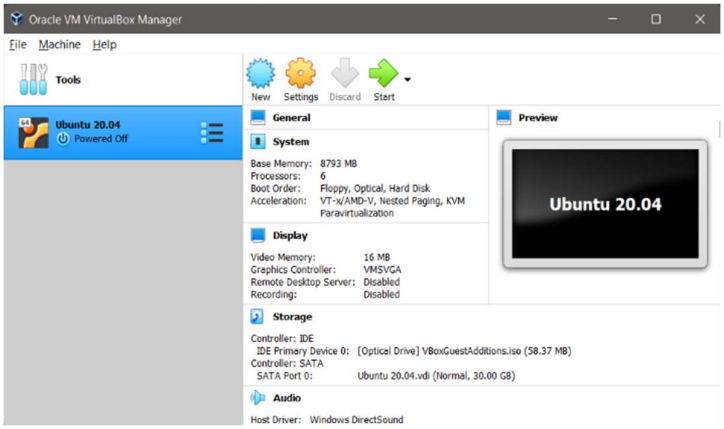


Fig 6.13 Click on New

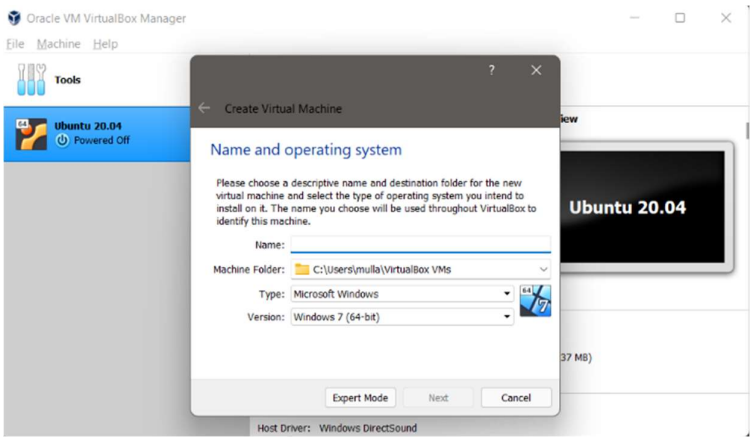


Fig 6.14 Create Virtual Machine Wizard

7. In 'Name' type name of the OS we are Installing. In our case type in Ubuntu 20.04.

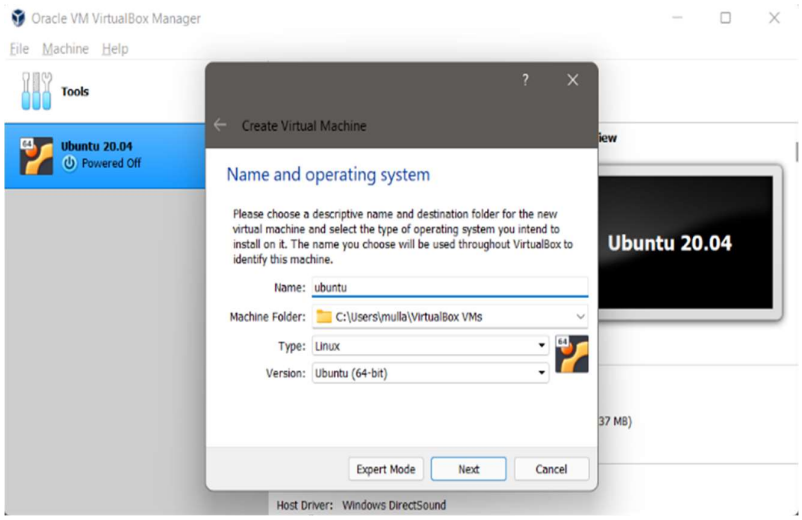


Fig 6.15 Type In The OS We Are Installing

8. Select Memory size as per our requirement.

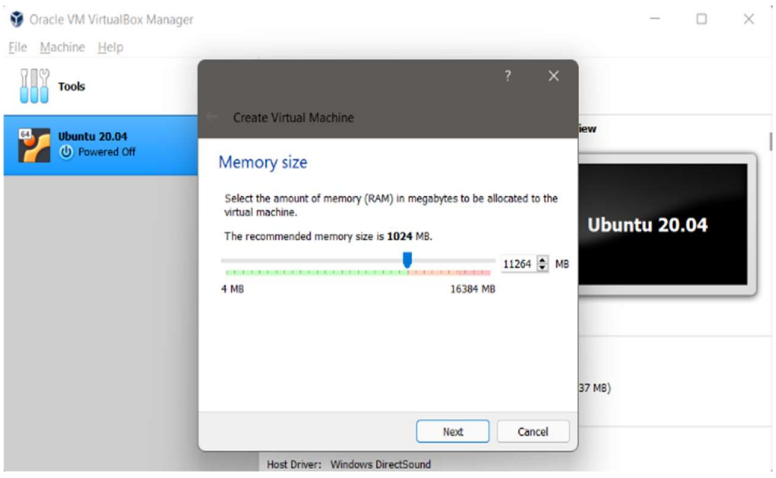


Fig 6.16 Select Memory Size

9. Select Hard disk Partition type and Hard Disk File Type, in our case we can create a virtual hard disk now

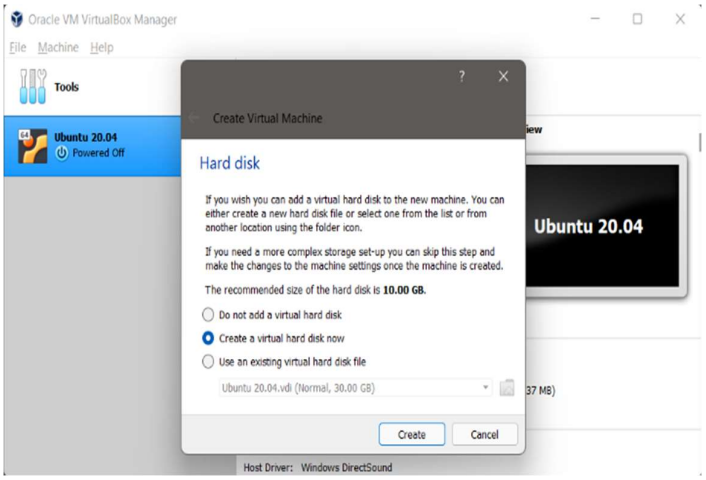


Fig 6.17 Select Hard disk Type

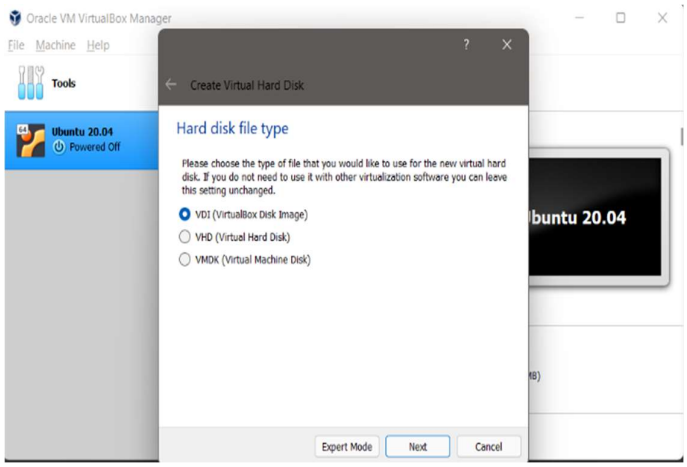


Fig 6.18 Selection of Hard Disk File Type

10. Select the storage as Dynamically allocated or Fixed Size, in our case we set in Fixed Size.

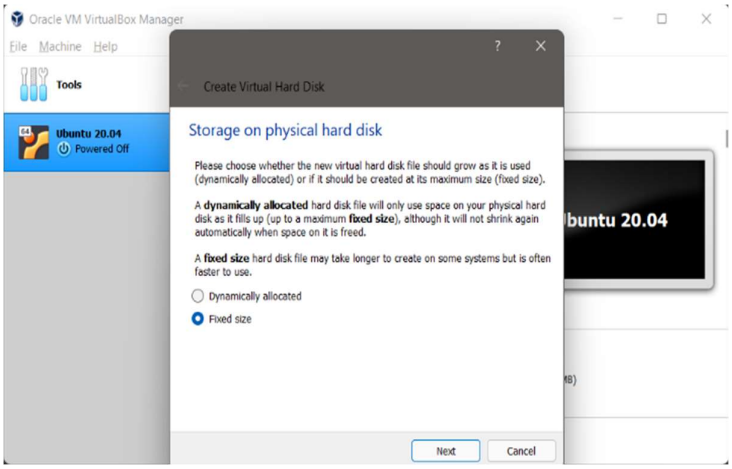


Fig 6.19 Storage Type Selection

11. Select the storage size, it should be above 10 GB and click on Create.

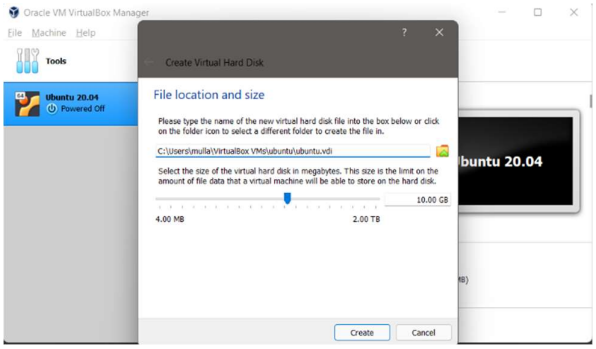


Fig 6.20 File Location And Size

12. A new Icon will be created with name Ubuntu 20.04, select that and click on Settings->General->Select Bidirectional in Shared Clipboard and Drag'n'Drop.

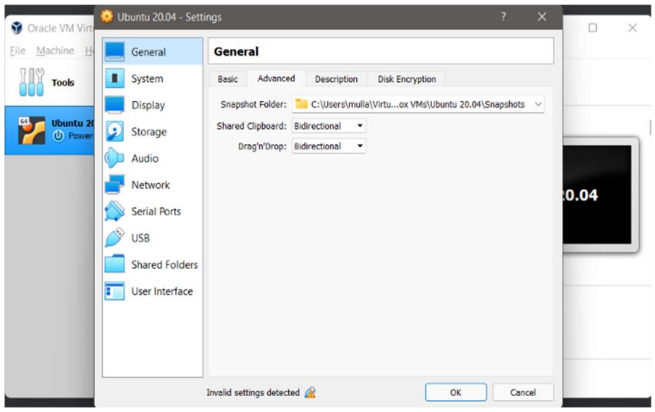


Fig 6.21 Ubuntu VM Settings

13. Go to System, under motherboard select the base memory, in our case we have chosen 8GB.

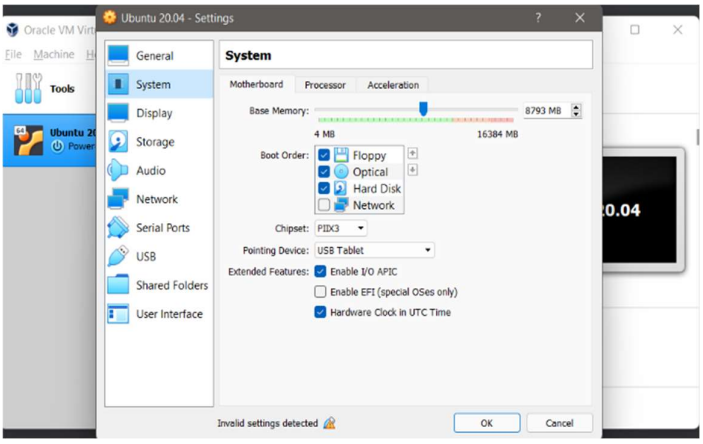


Fig 6.22 System Settings Configuration

14. Under system go to Processor, select the number of processing cores needed, in our case we have chosen 6 cores.

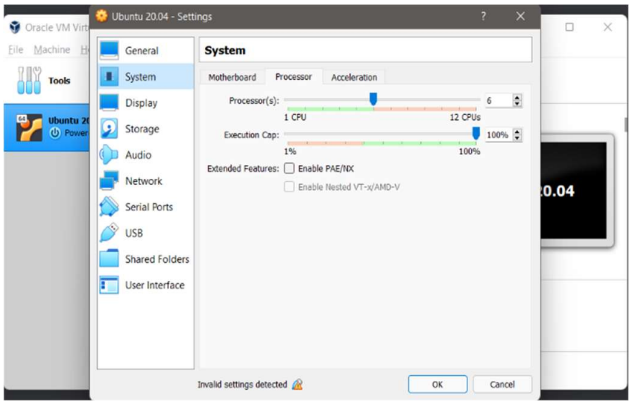


Fig 6.23 Processor Configuration

15. Go to Storage , point the location where the Ubuntu_20.04.iso file is stored. Click on OK and every settings will be saved.

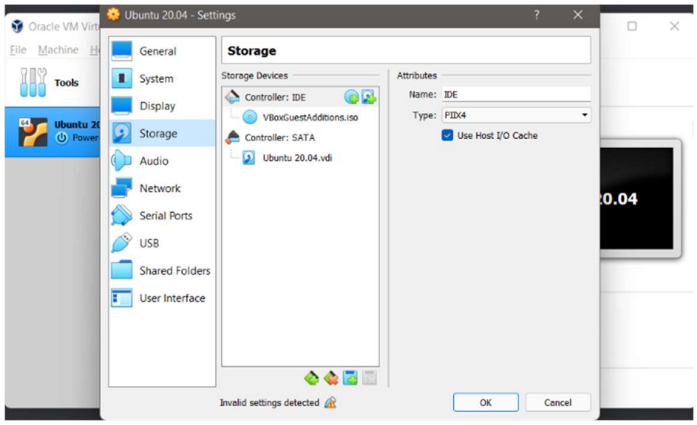


Fig 6.24 Storage Configuration

16. Now select the Ubuntu 20.04 Virtual Machine and Click on Start, this will open a new window with Ubuntu OS installation setup.

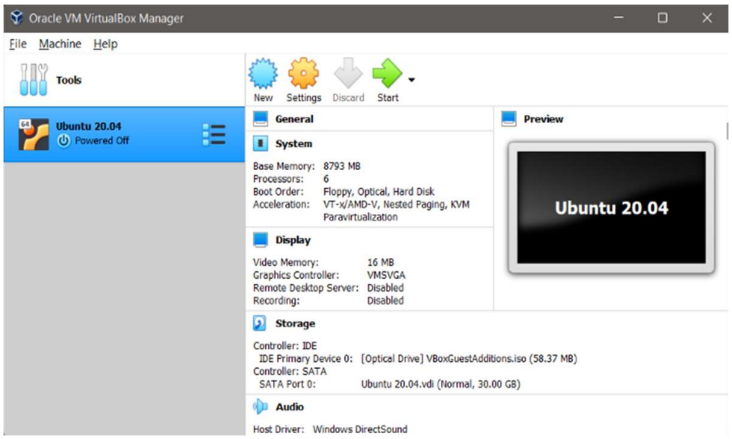


Fig 6.25 System Specification

17. After Booting this screen will appear.

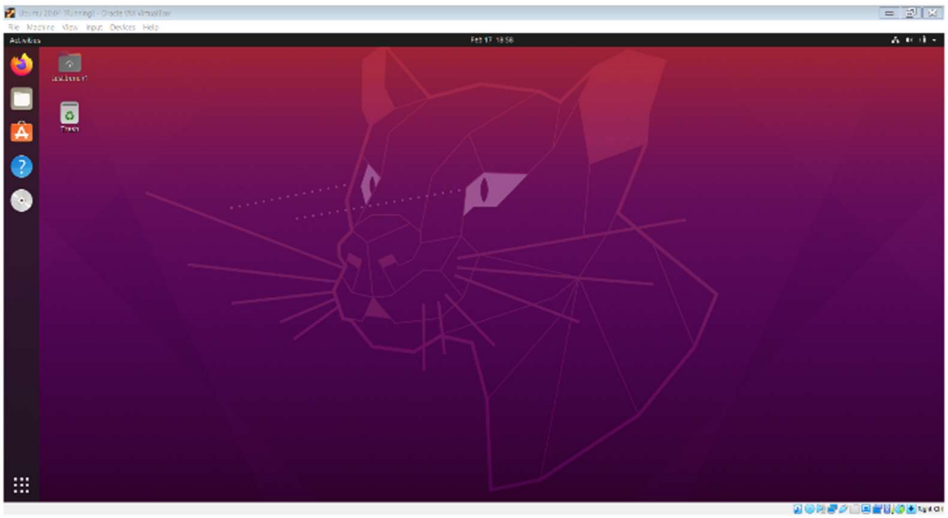


Fig 6.26 Ubuntu Desktop

6.1.2.2 INSTALLING ROS:

1. After opening the Ubuntu Virtual Box Successfully, check for updates and install them.

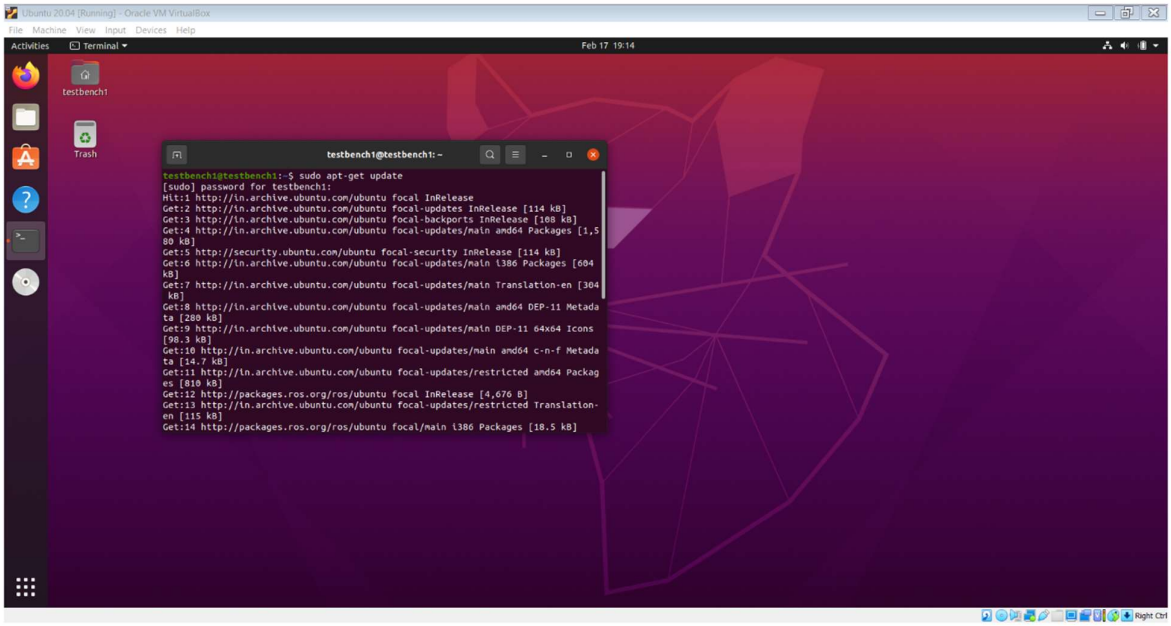


Fig 6.27 Ros Installation Via Terminal

2. Commands to get and install update:

```
$ sudo apt-get update
```

```
$ sudo apt-get upgrade
```

3. To install ROS open a new Terminal and enter the code given below:

```
$ wgethttps://raw.githubusercontent.com/ROBOTIS-GIT/robotis_tools/master/install_ros_noetic.sh

$ chmod 755 ./install_ros_noetic.sh

$ bash ./install_ros_noetic.sh
```

This will install ROS Noetic Packages.

4. To install the Dependent ROS Packages Enter the Commands given below:

```
$ sudo apt-get install ros-noetic-joy ros-noetic-teleop-twist-joy \
ros-noetic-teleop-twist-keyboard ros-noetic-laser-proc \
ros-noetic-rgbd-launch ros-noetic-rosserial-arduino \
ros-noetic-rosserial-python ros-noetic-rosserial-client \
ros-noetic-rosserial-msgs ros-noetic-amcl ros-noetic-map-server \
ros-noetic-move-base ros-noetic-urdf ros-noetic-xacro \
ros-noetic-compressed-image-transport ros-noetic-rqt* ros-noetic-rviz\
ros-noetic-gmapping ros-noetic-navigation ros-noetic-interactive-markers
```

Now ROS Noetic with every libraries needed is installed successfully.

5. To test open a new Terminal and enter “roscore” and press enter. The result should be “started core service”.

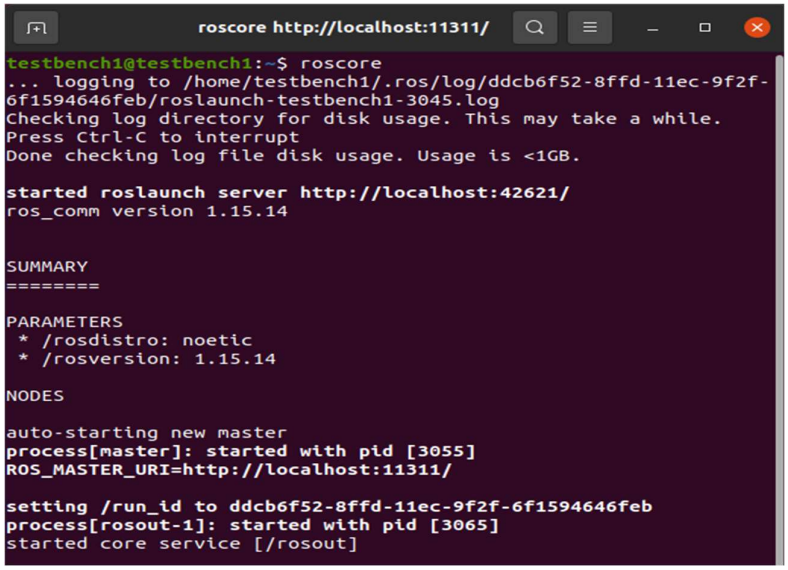


Fig 6.28 Initializing Ros

6.2 DRIVER INSTALLATION:

In Windows just download the hls_lfcd_lds_driver.zip file and paste in the ROS folder.

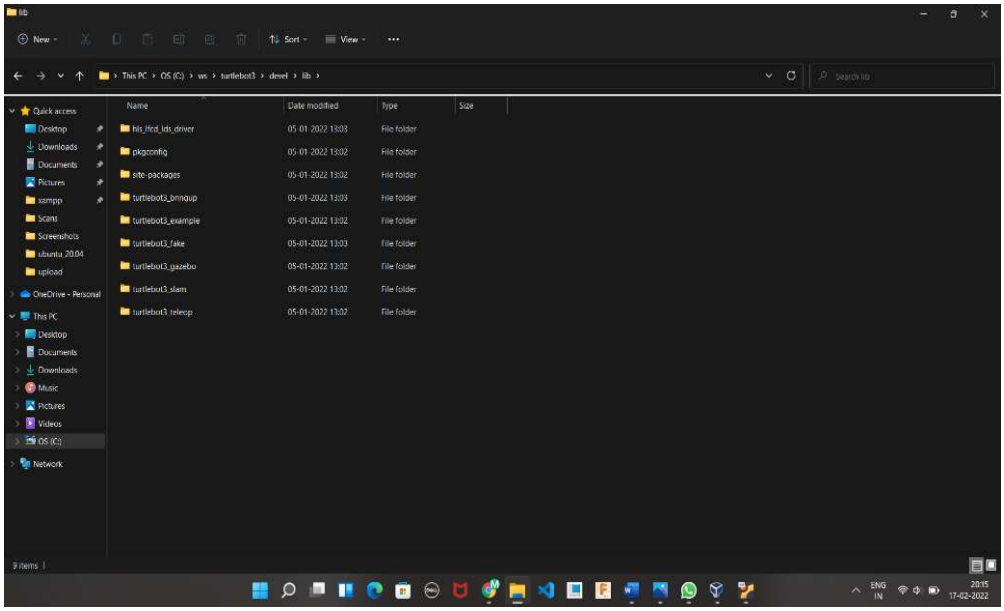


Fig 6.29 Driver Installation In Windows

In Linux Machine, Open Terminal and enter the commands given below:

$ sudo apt-get install ros-kinetic-hls-lfcd-lds-driver

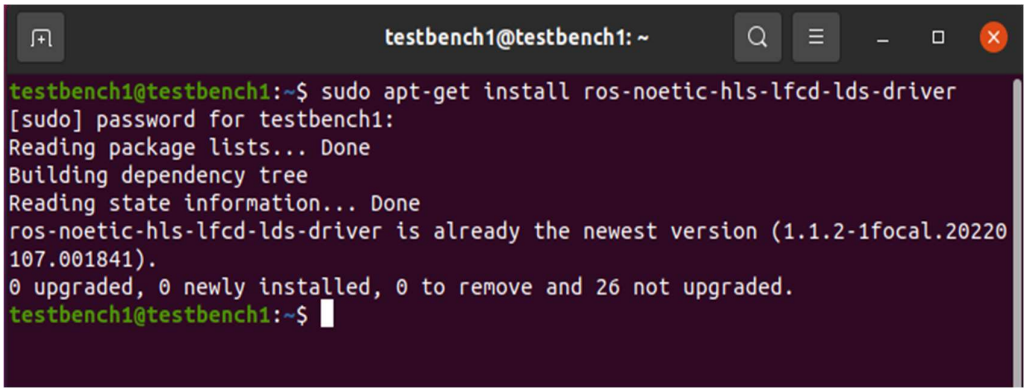


Fig 6.30 Driver Installation In Linux

To set Permission for LDS-01, enter the command given below:

$ sudo chmod a+rw /dev/ttyUSB0

Now the Drivers for LDS-01 Lidar is successfully installed.

CHAPTER 7
FABRICATION

CHAPTER 7

FABRICATION

7.1 BLOCK DIAGRAM:

All the systems are interlinked together during the fabrication stage. The various systems in a AGV are:

- Motor Drive System
- Control System
- Sensor System
- Processing Unit
- Communication System
- Power Distribution System
- Charging/Discharging System

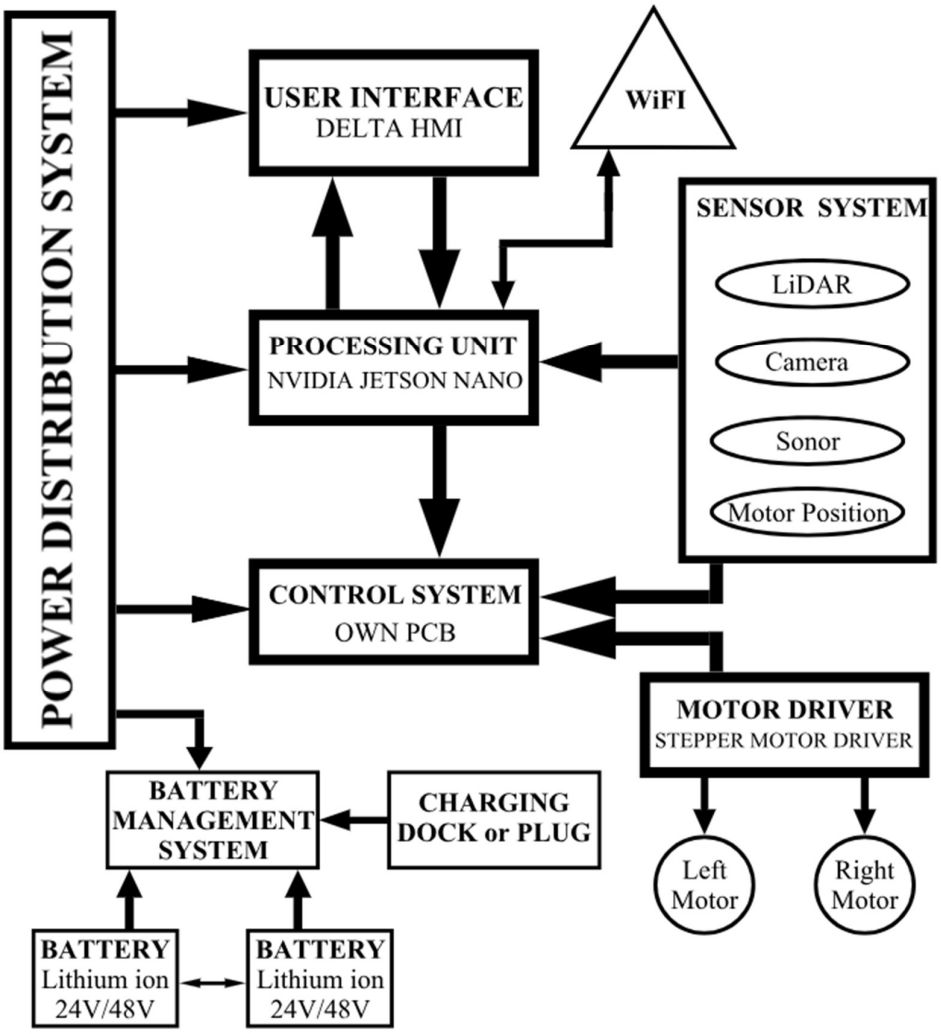


Fig 7.1 Block Diagram of AGV Working.

7.2 STRESS TEST:

The construction of AGV starts with the building the Frame structure, using 3D Solid Modelling software the frame is designed. This designed framed is then undergoes stress and strain test.

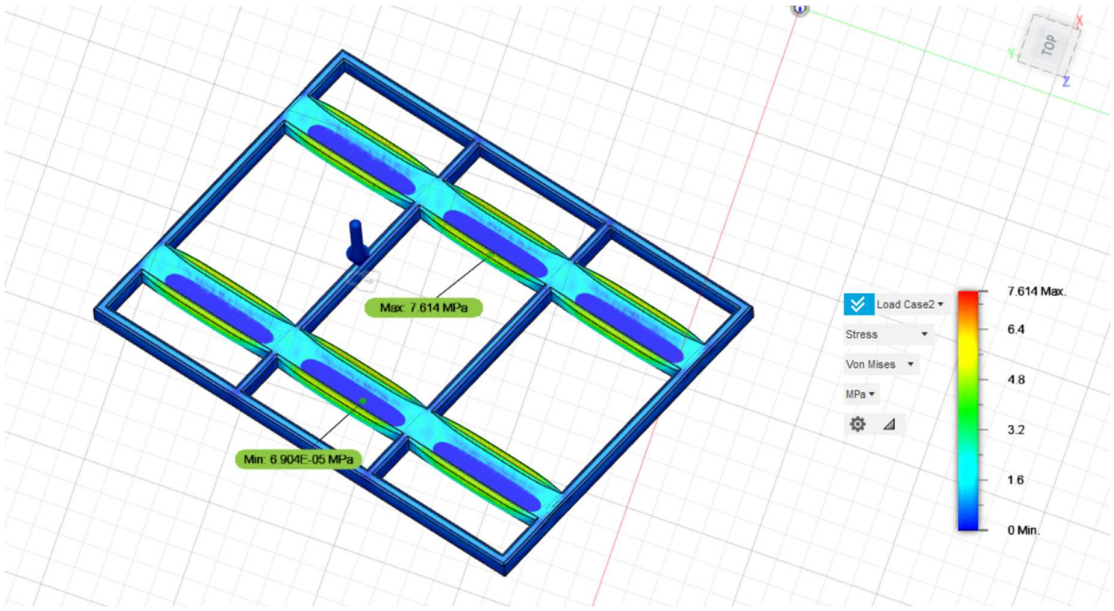


Fig 7.2 Stress test for 500 kgf.

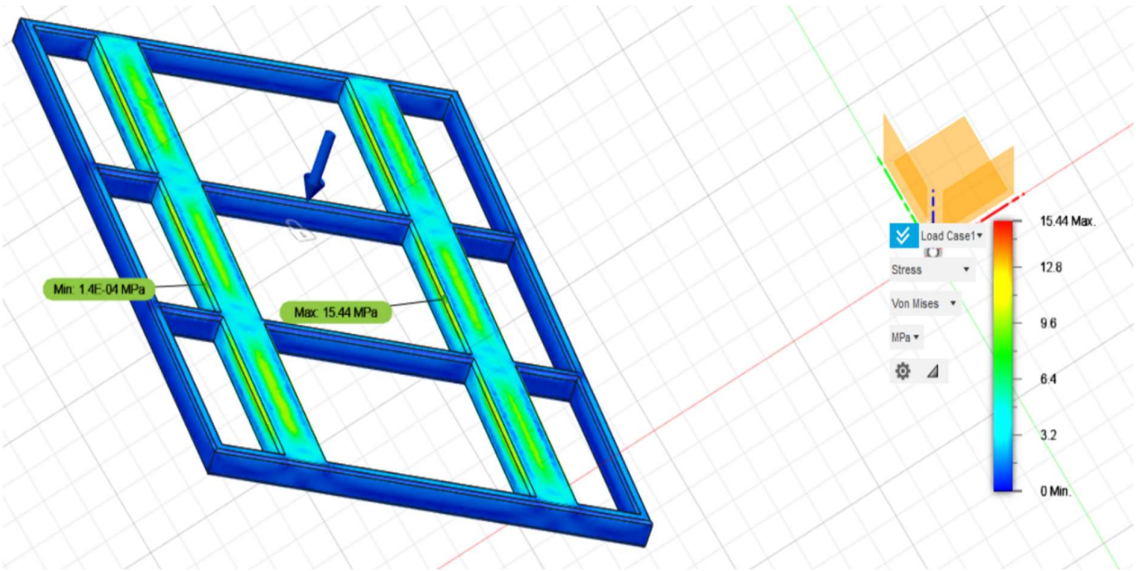


Fig 7.3 Stress test for 1000 kgf.

Now the full structure is built using Mild Steel and then holes are drilled onto it for mounting the hardware. First the supporting wheels are mounted onto. Then the main drive system is mounted.

7.3 CONSTRUCTION PROCESS:

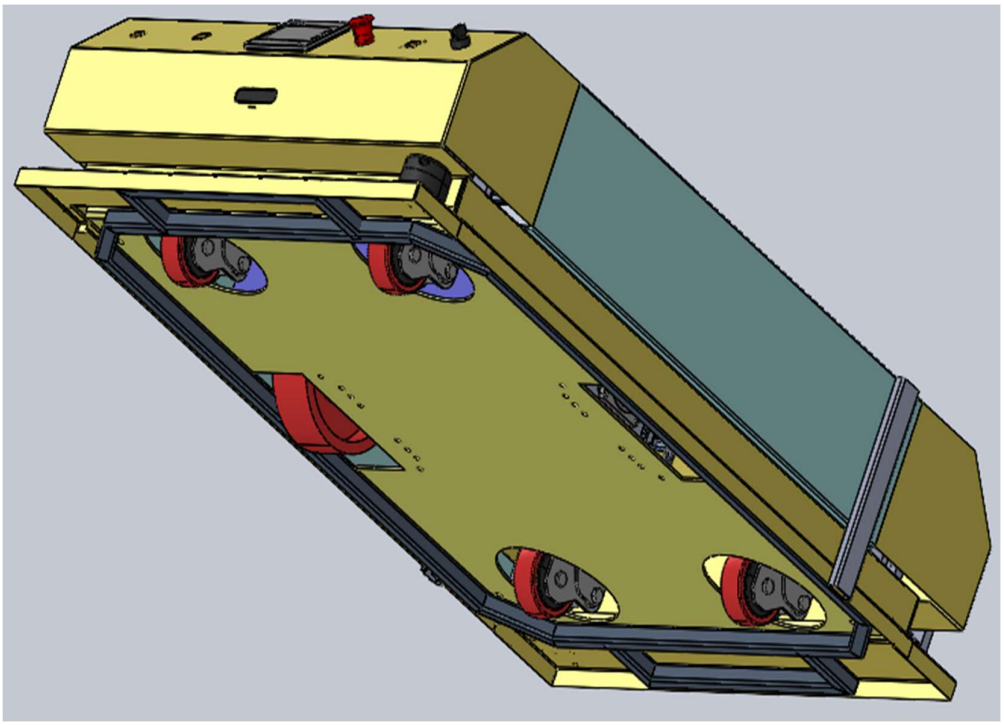


Fig 7.4 Bottom View of the AGV.

Then the sensors are mounted the LiDAR, Intel Real Sense Camera, Distance sensor using mounting brackets and covers. They are tightly sealed in place to avoid movements.

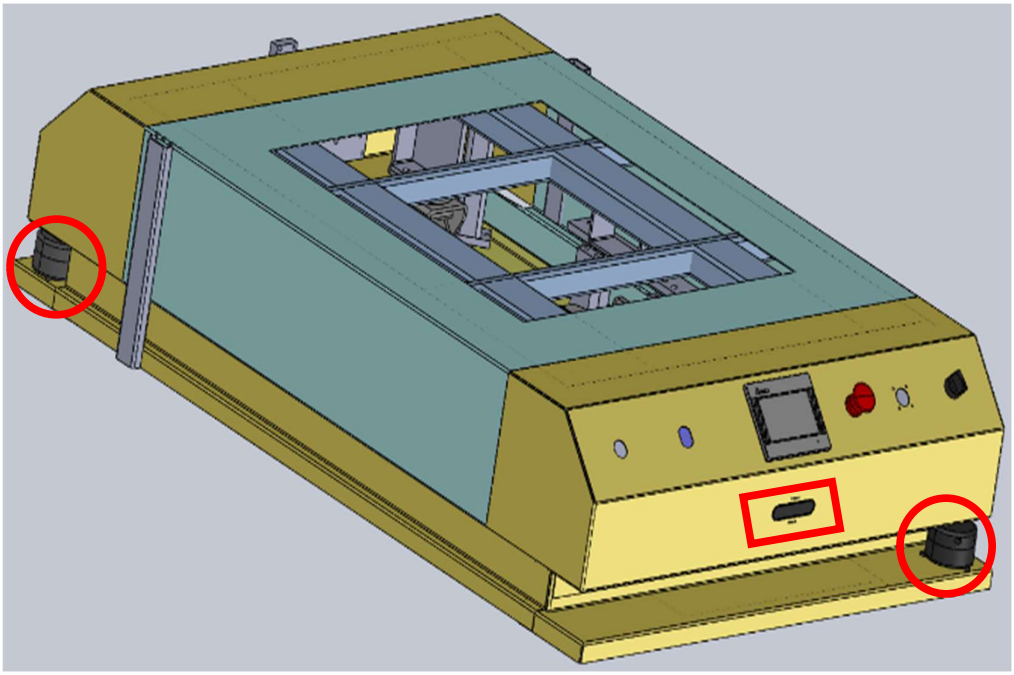


Fig 7.5 Sensors mounted in AGV

Then the Controlling unit the Nvidia Jetson Nano Development board is fixed and necessary connections are made. For user interface the Delta HMI is also Mounted, this also does some processing and interconnects with the other PLC devices in and around the Shop Floor.

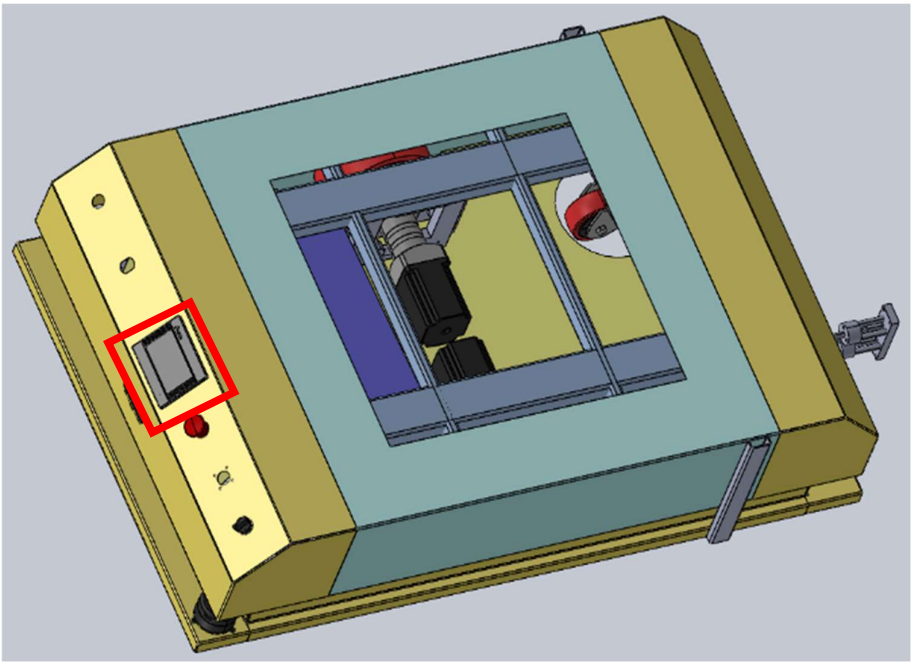


Fig 7.6 Isometric View of AGV

We also have a Emergency Stop switch to stop the AGV during malfunction and updates. It is mounted in the front, near to the HMI screen.

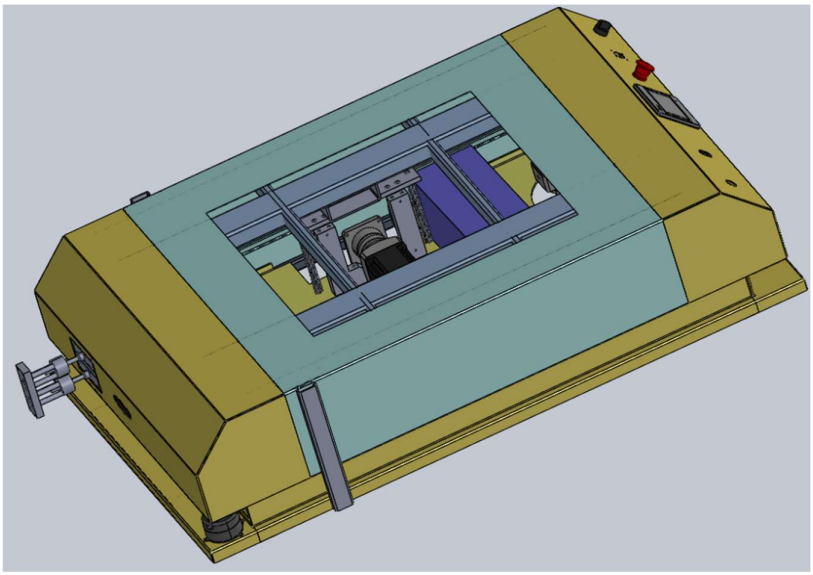


Fig 7.7 Isometric view from Behind

Then the battery and the battery management system are fixed. The battery is placed in the Front Centre to balance the weight and distribute it equally.

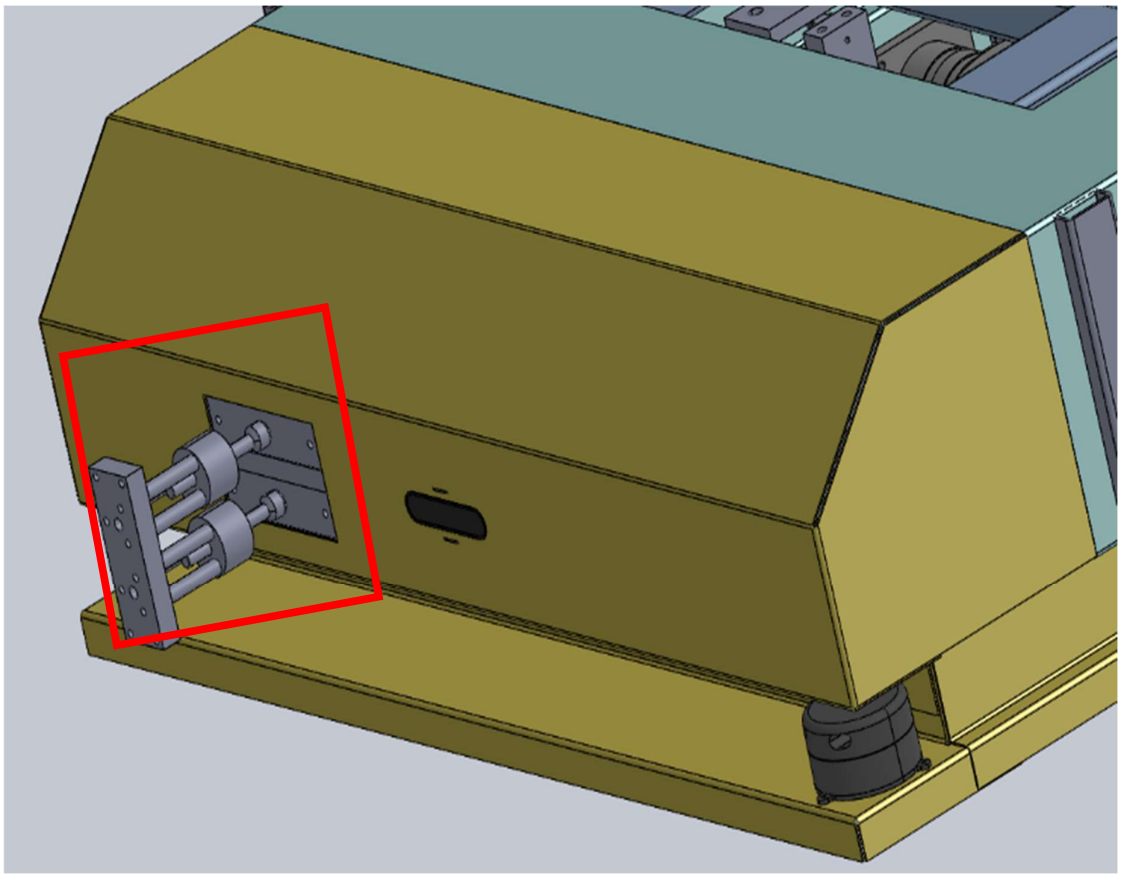


Fig 7.8 Charging Dock Plug

Then the Charging Dock module is fixed and ensured that is not touched to the body of the AGV.

Now the complete AGV is Covered with Sheet Metal covers they are strong enough to handle the wear and tear applied over the body. Also, the Top plate for carrying the load it is Projected above few inches using supporting plates. The Top plate is Hard and Rigid.

7.4 FINAL PRODUCT:



Fig 7.9 Front view of Final product.



Fig 7.10 Isometric view of Final Product

CHAPTER 8
TESTING

CHAPTER 8

TESTING

8.1 COLLECTING DATA FROM LiDAR

To collect data from the lidar and display the live map in screen we have to configure the rviz file and saved in the .launch file format. This file contains the configuration lidar data with drivers installed for processing data.

8.1.1 IN WINDOWS

1. Open the ROS shortcut Command Prompt
2. Type roscore and click on enter, this initializes the ROS Noetic.

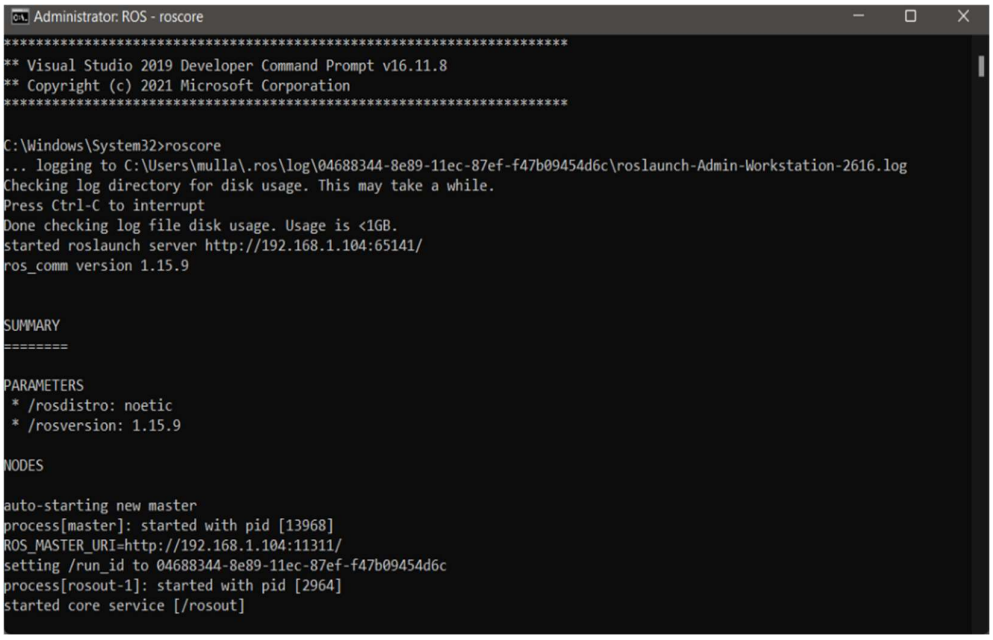


Fig 8.1 Initializing ROSCORE In Windows

3. Open another ROS Command Prompt.
4. Setup bash file first, enter the command given below:

$ c:\opt\ros\noetic\x64\setup.bat

$ c:\ws\turtlebot3\devel\setup.bat

To run hlds_lds_publisher Node, type in the command given below:

roslaunch hls_lfcd_lds_driver hlds_laser.launch.

This just generate coordinates.

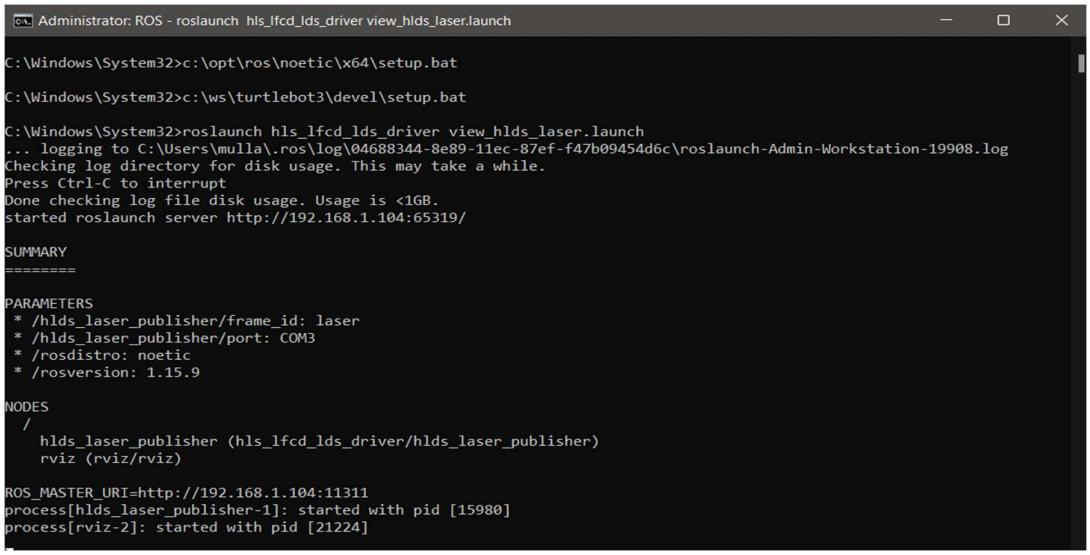


Fig 8.2 Launching Rviz File In Cmd

To run hlds_lds_publisher Node with rviz to see the maping, type in the command given below:

```
roslaunch hls_lfcd_lds_driver view_hlds_laser.launch
```

7. Now RVIZ application will open with Displaying the LiDAR output:

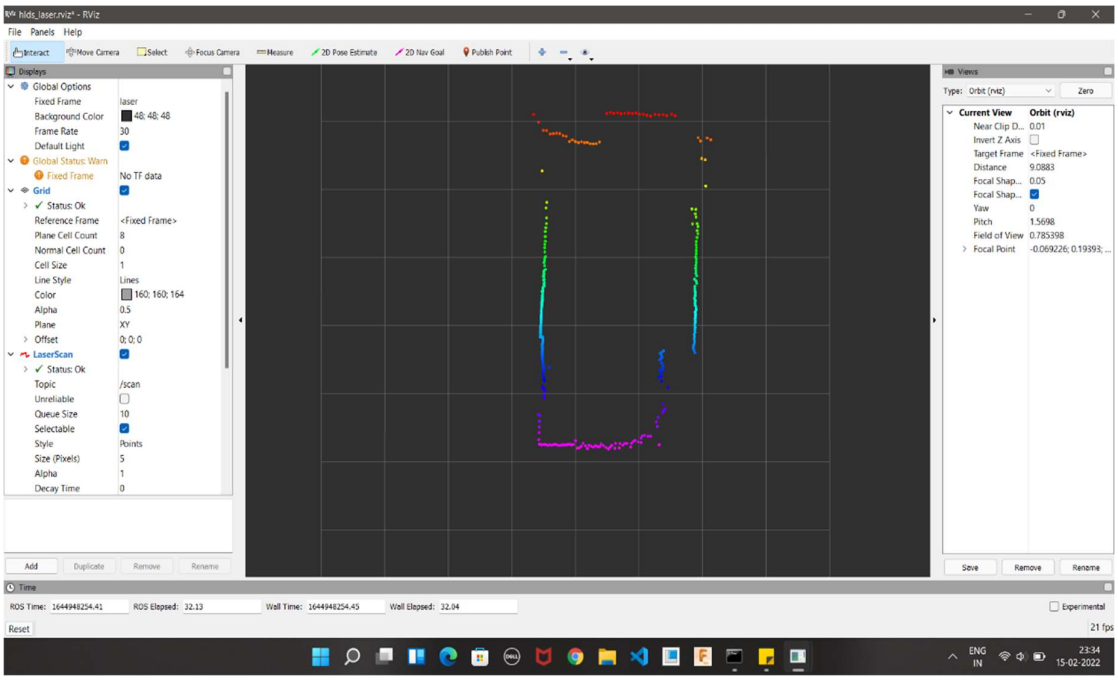


Fig 8.3 LIDAR Output In Rviz (Windows 11)

8.1.2 IN LINUX:

1. After launch virtual machine open a new terminal and launch ROS Noetic , type in “roscore” and click enter.

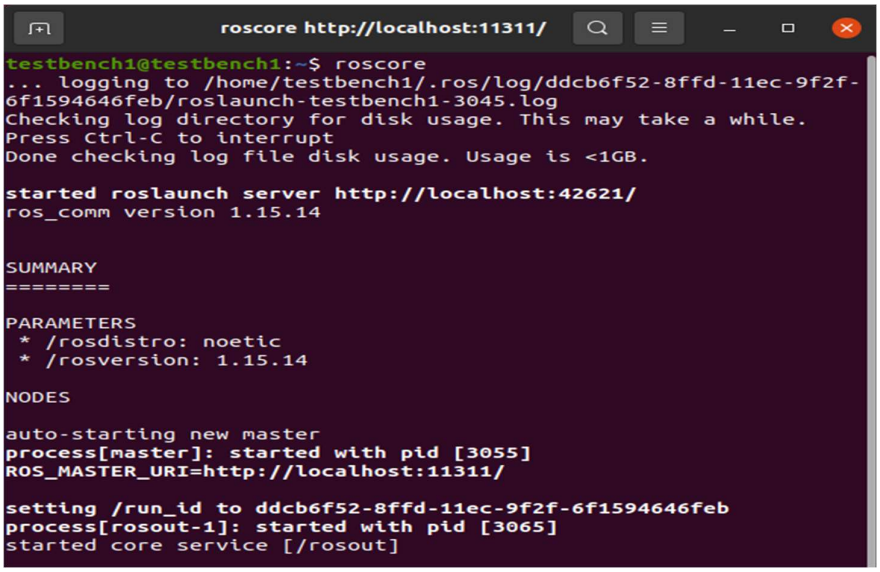


Fig 8.4 Roscore In Ubuntu Terminal

2. Now open a new Terminal and initialize connection and launch the hlds_lds_publisher Node with RVIZ , type in the command given below:

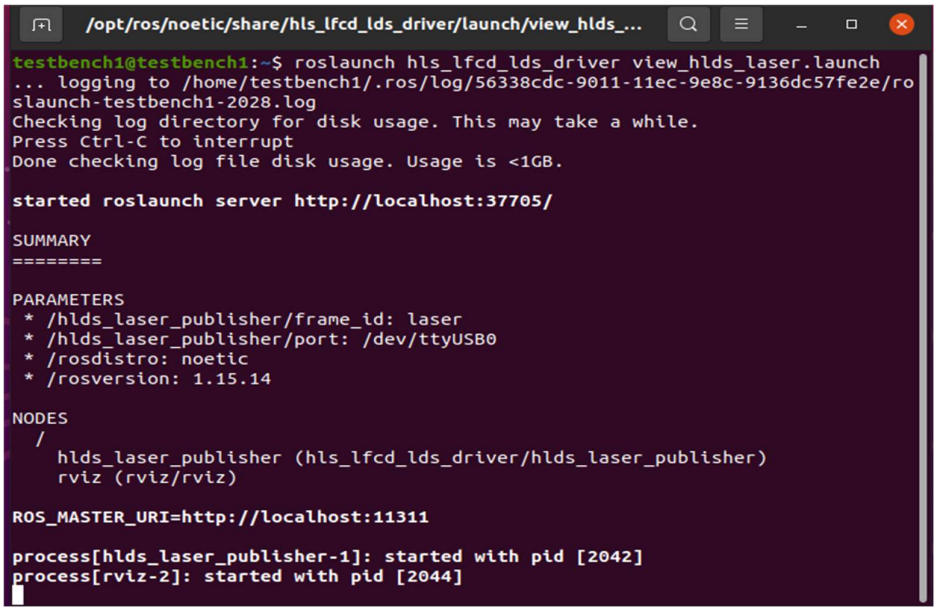


Fig 8.5 Launching Rviz File In Terminal

3. Now this will launch RVIZ application with LiDAR data and display the live map output.

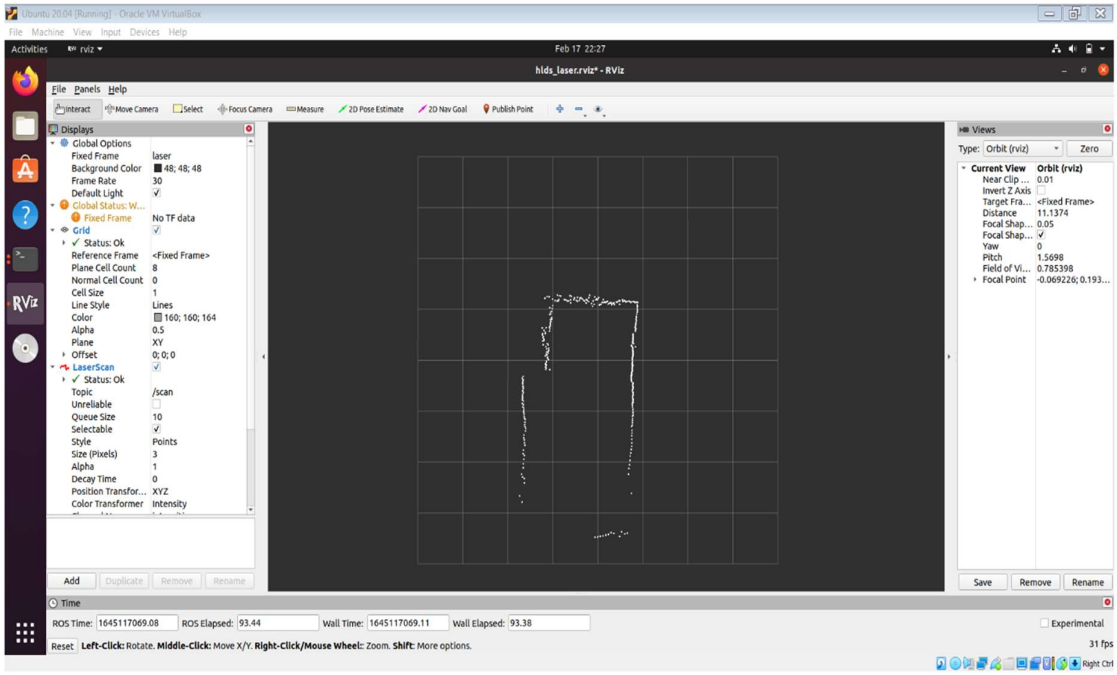


Fig 8.6 LIDAR Output In Rviz (Ubuntu20.04)

CONCLUSION

CONCLUSION

The AGV is a productivity increasing feature in a factory. During the manufacturing of this AGV we had found many of intelligence that can be given to it. We provide the functions like room mapping and collision avoiding. And the main function, transportation of goods from station to station. The followings are the main features of the prototype which we fabricated.

1. Speed of delivery
2. Adjustment of vehicle speed
3. Flexibility of path
4. Adaptive to changes in factory layouts
5. Avoid collision with other objects
6. Reduction in labour cost
7. Reduction in running cost compared to conveyer systems
8. Ability to add sensors to detect the payload conditions
9. Ability to adjust the lifting time
10. Continues cycle of working
11. Conditions for line following can be change easily

Automatic Guided Vehicle can be used in a wide variety of applications to transport many different types of material including pallets, rolls, racks, carts, and containers. AGVs excel in applications with the following characteristics:

- Repetitive movement of material over a distance
- Regular delivery of stable loads
- Medium throughput/volume
- When on time delivery is critical and late deliveries are causing inefficiencies
- Operation with at least two shifts
- Archive Systems
- Cross Docking
- Distribution

- High Density Storage
- High Speed Sortation
- Material Flow and Transport
- Production and Manufacturing Delivery Systems
- Production and Manufacturing Support Systems
- Warehouse Management and Control
- Work-In-Process Buffers

The fabricated models have following advantages while comparing with the existing models of this kind. The analysing of advantages helps to motivate the fabrication of AGV in the manufacturing industries. The important advantages are given below:

- Reduce manpower
- Increase productivity
- Eliminate unwanted fork trucks
- Reduce product damages
- Maintain better control of material management
- Traffic control is not needed in this system because of single carrier
- Suitable to transfer frames

Each of the machines has their own merits and demerits. During the production we had faced many problems. Much of them were solved during the assembling. But still some of them stand here, which can't have to remove. The followings are the limitations of the prototype fabricated:

- Installation cost is very high.
- AGVs are fragile and should be handled with care.
- Regular care, inspection and maintenance needed
- Should be recharged periodically
- AGV will stop delivery when it is forced off the path.
- Battery should be recharged during intervals.
- Sun light affect the movement.

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2. https://www.ros.org/
3. https://emanual.robotis.com/docs/en/platform/turtlebot3/appendix_lds_01/
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